BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Module 5 Packet.

Nervous Tissue · Brain · Spinal Cord · Peripheral · Autonomic

PART A	**Course Notes**	10 chapters, in the order the module teaches them. Your reference for every video, lab, and exam question in Module 5.
PART B	**Pre-work**	5 packets. Complete these by hand before the class day they belong to. These pages print landscape.
PART C	**Lab Structure List**	The Module 5 slice of the master lab list. Everything here is testable on the lab portion of the Module 5 exam.

Solano Community College · Fall 2026

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BIO 004 · MODULE 5

Contents

10 chapters, in the order the module teaches them. Part A of the Module 5 packet. Part B is the pre-work, Part C is the lab structure list.

Everything else for Module 5

Scan for the course home: the concept videos, Mastery OS, the calendar, and the exam schedule. Nothing behind this code is reprinted in this packet.

WHAT MODULE 5 COVERS

Module 5 is the nervous system, and it gets its own final. Forty three competencies, the same size as the whole of Module 3, in one system.

The ten chapters build outward from the cell. Nervous tissue is the material, and neuronal integration is the wiring logic that everything after it depends on. The brain and the brainstem are the central organ. The meninges and cerebrospinal fluid are what protect and float it, and the spinal cord is the cable running out of it. The peripheral chapters follow the wiring out to the body: spinal nerves and their plexuses, the twelve cranial nerves, and the autonomic system that runs underneath all of it without asking.

Read the chapters in order. Almost every question on this module is a pathway question or a lesion question, and both need the wiring in your head before the names will stick.

PART A

Course Notes.

10 chapters, in the order the module teaches them. Read a chapter before the class day it belongs to, then do the matching pre-work packet in Part B.

CHAPTER 1

Functional Organization and Nervous Tissue

FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE

Covers the organization of the nervous system and its divisions, the three functions, the structure of a neuron, the synapse, the classification of neurons, the neuroglia, myelination, and the collections of nervous tissue. The focus is on the structures and the job each one does.

BY THE END

1. Outline the organization of the nervous system and its divisions.
2. Identify the parts of a neuron and classify neurons by shape and by function.
3. Name the neuroglia and describe myelination and the collections of nervous tissue.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Nervous tissue

drsrennie-stack.github.io/new-build-bio4-solano/nervous-tissue-concept-videos.html

The Nervous System, an Overview

The nervous system gives the body the ability to feel, think, and act. It has two main divisions: the central nervous system and the peripheral nervous system.

THE TWO DIVISIONS AND THE STRUCTURES IN EACH

Structure	What it is
Nervous system	The body's fast control system; it lets us feel, think, and act
Central nervous system (CNS)	The brain and the spinal cord; it processes information and issues commands
Brain	The CNS organ in the cranial cavity, with about 85 billion neurons
Spinal cord	The CNS cord in the vertebral canal that links the brain to the body
Peripheral nervous system (PNS)	Every neural structure outside the brain and spinal cord
Nerves	Bundles of axons in the PNS; 12 pairs of cranial nerves and 31 pairs of spinal nerves
Ganglia	Small masses of neuron cell bodies located outside the CNS
Sensory receptors	Structures that monitor changes inside and outside the body

Divisions of the Peripheral Nervous System

The motor side of the peripheral nervous system is split by the kind of effector it controls. The somatic system runs skeletal muscle; the autonomic system runs the viscera; the enteric system runs the gut.

THE MOTOR DIVISIONS OF THE PNS

Structure	What it is
Somatic nervous system (SNS)	The voluntary division; its motor neurons carry commands from the CNS to skeletal muscle
Autonomic nervous system (ANS)	The involuntary division; its motor neurons carry commands to smooth muscle, cardiac muscle, and glands
Sympathetic division	The ANS division for fight or flight, supporting exercise and emergencies
Parasympathetic division	The ANS division for rest and digest, supporting routine activities
Enteric nervous system (ENS)	The brain of the gut, about one million neurons in the wall of the GI tract that can operate on their own

The Three Functions of the Nervous System

Whatever the task, the nervous system handles it in the same three steps. A stimulus comes in, the information is processed, and a response goes out.

1. **Sensory function** sensory receptors detect a change inside or outside the body and send the input toward the CNS
2. **Integrative function** the CNS analyzes the sensory input, stores it, and decides on a response
3. **Motor function** the CNS sends commands out to the effectors, the muscles and glands, which carry out the response

The Structure of a Neuron

A neuron is the signaling cell of the nervous system. It is excitable, it can carry an impulse, and it cannot divide. Every neuron has three basic parts.

THE PARTS OF A NEURON

Structure	What it is
Neuron	The excitable cell that receives, conducts, and transmits nerve impulses; neurons cannot divide
Cell body	The soma or perikaryon; it holds the nucleus and cytoplasm and is the metabolic center of the neuron
Dendrites	The branched, tree-like processes that receive incoming signals
Axon	The single long process that carries the nerve impulse away toward other cells
Nissl bodies	Clusters of rough ER and ribosomes in the cell body; the site of protein synthesis
Neurofibrils	Cytoskeletal fibers that give the neuron its shape and support
Microtubules	Cytoskeletal tubes that transport materials along the neuron
Axon hillock	The cone-shaped region where the cell body joins the axon
Axoplasm and axolemma	The cytoplasm of the axon and the plasma membrane of the axon
Axon terminals	The fine branches at the far end of the axon that contact the target cell
Lipofuscin	A harmless pigment that builds up in neurons with age

The Synapse

A synapse is the junction where a neuron passes its signal to another neuron, a muscle, or a gland. At a chemical synapse the signal crosses as a chemical message.

THE SYNAPSE AND ITS PARTS

Structure	What it is
Synapse	The site where a neuron communicates with another neuron, a muscle cell, or a gland
Axodendritic synapse	An axon ending on a dendrite
Axosomatic synapse	An axon ending on a cell body
Axoaxonic synapse	An axon ending on another axon
Chemical synapse	A synapse with no direct contact; the signal crosses as neurotransmitter molecules
Synaptic cleft	The narrow gap between the two cells that the neurotransmitter diffuses across
Synaptic end bulbs	The swellings at the axon terminal that hold the signal-sending machinery
Synaptic vesicles	The tiny sacs inside the end bulbs that store neurotransmitter
Neurotransmitters	The chemical messengers that carry the signal across the cleft to the next cell

Classification of Neurons

Neurons are classified two ways: by their shape, which is the structural classification, and by the direction they carry signals, which is the functional classification.

Structural classification, by shape

NEURONS NAMED FOR THE NUMBER OF PROCESSES

Structure	What it is
Multipolar neuron	Many dendrites and one axon; the most common type, found in the brain and cord and as motor neurons
Bipolar neuron	One dendrite and one axon; found in the retina, the inner ear, and the olfactory area
Unipolar neuron	A single fused process; the typical shape of a sensory neuron
Purkinje cells	A distinctive multipolar neuron found in the cerebellum
Pyramidal cells	A distinctive multipolar neuron found in the cerebral cortex

Functional classification, by direction

NEURONS NAMED FOR THE DIRECTION THEY CARRY A SIGNAL

Structure	What it is
Sensory (afferent) neurons	Carry sensory input toward the CNS; most are unipolar
Motor (efferent) neurons	Carry motor commands away from the CNS; these are multipolar
Interneurons	Also called association neurons; they connect sensory and motor neurons within the CNS and are multipolar

Neuroglia

Neuroglia, or glial cells, are the support cells of nervous tissue. They are smaller and far more numerous than neurons, and unlike neurons they can still divide.

WHAT THE GLIAL CELLS ARE

Structure	What it is
Neuroglia	The support cells of nervous tissue; they nourish, support, and protect neurons and make up about half the volume of the CNS

Glia of the central nervous system

THE FOUR GLIA OF THE CNS AND WHAT EACH ONE DOES

Glial cell	Role
Astrocytes	Star-shaped cells that give structural support, help form the blood-brain barrier, guide neuron growth, and maintain the chemical environment
Oligodendrocytes	Cells that form the myelin sheaths of the CNS
Microglia	Small phagocytic cells that remove debris and microbes
Ependymal cells	Cells that line the ventricles and central canal and help produce and circulate cerebrospinal fluid

Glia of the peripheral nervous system

THE TWO GLIA OF THE PNS AND WHAT EACH ONE DOES

Glial cell	Role
Schwann cells	Cells that form the myelin sheath around PNS axons and aid axon regeneration
Satellite cells	Cells that surround and support the cell bodies of PNS neurons in ganglia

Myelination

Myelin is a fatty wrapping around an axon. It insulates the axon and makes the nerve impulse travel much faster.

THE MYELIN SHEATH AND WHAT IT DOES

Structure	What it is
Myelin sheath	A multilayered covering of lipid and protein wrapped around an axon
Function of myelin	It insulates the axon and speeds up the nerve impulse
Nodes of Ranvier	The gaps in the myelin sheath, spaced along the axon
Saltatory conduction	The fast conduction in which the impulse jumps from node to node

The cell that builds the myelin differs between the PNS and the CNS, and that difference decides whether a damaged axon can regrow. Compare them.

PNS AND CNS MYELINATION COMPARED

Feature	PNS myelination	CNS myelination
Cell that makes the myelin	A Schwann cell, one per axon segment	An oligodendrocyte, one cell for up to 15 axons
Neurolemma	Present; it is the outer layer of the Schwann cell	Absent
Axon regeneration	Possible; the neurolemma guides the axon as it regrows	Very limited, because there is no neurolemma

Myelination increases from birth to maturity, so an infant's nerve conduction is slower than an adult's.

HOLD ONTO THIS

One Schwann cell myelinates one segment of one axon and leaves a neurolemma behind. One oligodendrocyte reaches up to 15 axons and leaves none. That single difference is why a cut nerve in the arm can regrow and a cut tract in the cord usually cannot.

Collections of Nervous Tissue

The same kind of structure has a different name depending on whether it sits in the CNS or the PNS. Learn these as pairs.

THE SAME STRUCTURE UNDER TWO NAMES

Structure	In the CNS	In the PNS
A cluster of cell bodies	Nucleus	Ganglion
A bundle of axons	Tract	Nerve

WHITE MATTER AND GRAY MATTER, AND WHERE EACH ONE SITS

Structure	What it is
White matter	Nervous tissue made of myelinated axons
Gray matter	Nervous tissue made of cell bodies, dendrites, unmyelinated axons, and neuroglia
In the spinal cord	White matter is on the outside, gray matter is on the inside
In the brain	Gray matter is on the outside, white matter is on the inside

HOLD ONTO THIS

Gray is where the cell bodies are, white is where the myelin is. The two swap positions between the cord and the brain, so name the organ before you name the layer.

WHEN THE SUPPORT CELLS FAIL

Neurons cannot divide, so a tumor of nervous tissue almost never comes from neurons. It comes from the **glia**, which can still divide. A **glioma** is a tumor of glial cells, and it is the most common type of brain tumor.

Myelin is as important as the axon it wraps. In **multiple sclerosis**, the immune system strips myelin from CNS axons and signals slow or fail. The presence of the **neurolemma** on Schwann cells, and its absence on oligodendrocytes, is why a cut nerve in the arm can regrow but a cut tract in the cord usually cannot.

The **cell body** is the neuron's life-support. Because the axon depends on the cell body for proteins, an injury that separates the two kills the axon beyond the cut. The location of a nerve injury, near or far from the cell body, helps predict whether the neuron survives.

CHAPTER 2

Gross Anatomy & Neuronal Integration

GROSS ANATOMY AND NEURONAL INTEGRATION

Covers gray and white matter, the names for bundles of fibers and clusters of cell bodies, the reflex arc, types of reflexes, and neuronal pools.

BY THE END

1. Distinguish gray matter from white matter.
2. Define the terms tract, nerve, nucleus, and ganglion.
3. Name the five components of a reflex arc in order.
4. Compare the types of reflexes and describe a neuronal pool.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Neuronal integration

drsrennie-stack.github.io/new-build-bio4-solano/nervous-tissue-concept-videos.html

An Overview

Covers two things: how nervous tissue is organized into the larger structures of the system, and how neurons work together in groups to process information.

THE TERMS THIS PAGE BUILDS ON

Structure	What it is
Gross organization	How nervous tissue is arranged into the brain, cord, and nerves
Neuronal integration	How neurons work together in groups to receive, combine, and respond to information
Gray matter	Regions made mostly of neuron cell bodies, dendrites, and synapses
White matter	Regions made mostly of myelinated axons
Reflex	A fast, automatic, predictable response to a stimulus

Gray Matter and White Matter

Nervous tissue sorts into two looks. Where cell bodies gather, the tissue looks gray; where myelinated axons run, it looks white. Compare them.

GRAY MATTER AND WHITE MATTER COMPARED

Type	What it is made of	Where it is
Gray matter	Neuron cell bodies, dendrites, and unmyelinated fibers	The cortex of the brain and cerebellum, and the inner core of the spinal cord
White matter	Myelinated axons running in bundles	Beneath the cortex of the brain, and the outer part of the spinal cord

Bundles of Fibers and Clusters of Cell Bodies

The same structure has a different name in the CNS and the PNS. Learn the four terms as two pairs: bundles of axons, and clusters of cell bodies.

THE FOUR TERMS, WHERE EACH ONE BELONGS, AND WHAT IT IS

Term	Where	What it is
Tract	CNS	A bundle of axons within the central nervous system
Nerve	PNS	A bundle of axons within the peripheral nervous system
Nucleus	CNS	A cluster of neuron cell bodies within the central nervous system
Ganglion	PNS	A cluster of neuron cell bodies within the peripheral nervous system

HOLD ONTO THIS

Two of the four terms are bundles of axons, tract and nerve, and two are clusters of cell bodies, nucleus and ganglion. Decide which kind of structure you are looking at first, then decide whether you are inside the CNS or outside it.

The Reflex Arc

A reflex is fast because it runs a short, fixed pathway. Every reflex arc has the same five components, in order.

1. **Receptor** detects the stimulus
2. **Sensory neuron** carries the signal from the receptor to the CNS
3. **Integration center** one or more synapses in the CNS where the signal is processed
4. **Motor neuron** carries the command from the CNS to the effector
5. **Effector** the muscle or gland that carries out the response

Types of Reflexes

Reflexes are sorted several ways: by the effector, by the number of synapses, and by whether they are inborn. Compare them.

THE TYPES OF REFLEXES AND WHAT EACH NAME IS BASED ON

Reflex type	Basis of the name	Example
Somatic reflex	The effector is skeletal muscle	The patellar, or knee-jerk, reflex
Autonomic reflex	The effector is smooth muscle, cardiac muscle, or a gland	The pupillary light reflex
Monosynaptic reflex	Just one synapse in the integration center	The stretch reflex
Polysynaptic reflex	Two or more synapses, with interneurons	The withdrawal reflex
Innate reflex	Present from birth, not learned	The blink reflex
Learned reflex	Built through practice and repetition	Catching a suddenly dropped object

Neuronal Pools and Circuits

Neurons rarely act alone. They are wired into groups, the neuronal pools, and the wiring pattern shapes how a signal is handled.

THE POOLS AND THE CIRCUIT PATTERNS

Structure	What it is
Neuronal pool	A functional group of interconnected neurons in the CNS that processes information together
Convergence	A circuit in which many neurons feed their signals onto one neuron
Divergence	A circuit in which one neuron spreads its signal to many neurons
Serial processing	A signal passing along a single straight chain of neurons, as in a reflex arc
Parallel processing	A signal sent along several pathways at the same time
Integration	The combining of all the incoming signals so that a neuron produces a single response

Sensory and Motor Pathways

Information travels the CNS on long tracts: sensory signals climb upward, motor commands descend, and most pathways cross to the opposite side.

RECEPTORS, PATHWAYS, AND CROSSING

Structure	What it is
Sensory receptor	A structure that detects a stimulus and triggers a signal in a sensory neuron
Types of receptors	Named for what they detect: mechanoreceptors, thermoreceptors, photoreceptors, chemoreceptors, and nociceptors for pain
Ascending pathways	The tracts that carry sensory information up the spinal cord to the brain
Descending pathways	The tracts that carry motor commands down from the brain to the body
Decussation	The crossing of a pathway to the opposite side, so each half of the brain serves the opposite half of the body

READING THE WIRING IN THE CLINIC

A tap below the kneecap triggers the **patellar reflex**, and a clinician watches the leg kick. That single test checks the whole reflex arc: the receptor, the sensory and motor neurons, and the spinal cord in between. A missing reflex points to a break somewhere along that pathway.

Because most pathways **decussate**, crossing to the opposite side, the right half of the brain controls and senses the left half of the body. This is why a stroke on one side of the brain causes weakness on the opposite side.

The split into **gray** and **white matter** shows up on a brain or spinal cord scan, and a change in that pattern, gray where white belongs or the reverse, is one of the things a radiologist looks for.

CHAPTER 3

The Brain

THE CENTRAL NERVOUS SYSTEM, THE BRAIN

Covers the development of the brain, the cerebrum with its lobes, white matter tracts, and basal ganglia, the functional areas of the cortex, the diencephalon, the cerebellum, the limbic system, and the arterial supply of the brain. The focus is on the structures and the job each one does.

BY THE END

1. Trace the development of the brain from the neural tube vesicles.
2. Identify the cerebrum, its lobes, white matter tracts, and basal ganglia.
3. Map the functional areas of the cerebral cortex.
4. Name the parts of the diencephalon, the cerebellum, the limbic system, and the arteries that supply the brain.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The brain

drsrennie-stack.github.io/new-build-bio4-solano/brain-concept-videos.html

The Brain, an Overview

The brain sits within the cranial cavity and holds roughly 85 billion neurons. It is the body's master control and integration center. The brain has four major parts.

THE FOUR MAJOR PARTS, AND THE TWO KINDS OF NERVOUS TISSUE

Structure	What it is
Brain	The organ within the cranial cavity, the control and integration center of the body
Cerebrum	The largest brain part, the seat of conscious thought, sensation, and voluntary movement
Diencephalon	The deep brain region between the cerebrum and the brainstem
Brainstem	The stalk connecting the brain to the spinal cord, covered on its own sheet
Cerebellum	The structure behind the brainstem that coordinates movement
Gray matter	Neuron cell bodies, dendrites, and unmyelinated fibers; on the outside of the brain
White matter	Bundles of myelinated axons; on the inside of the brain

Development of the Brain

The brain and spinal cord develop from the ectodermal neural tube. The tube swells at its anterior end into a series of vesicles that mature into the adult brain.

FROM TUBE TO VESICLES

Structure	What it is
Neural tube	The ectodermal tube from which the entire central nervous system develops
Primary brain vesicles	The first three swellings: the prosencephalon (forebrain), the mesencephalon (midbrain), and the rhombencephalon (hindbrain)
Secondary brain vesicles	The prosencephalon divides into the telencephalon and diencephalon; the rhombencephalon divides into the metencephalon and myelencephalon

Each secondary vesicle matures into specific adult structures. Compare them.

SECONDARY VESICLES AND WHAT EACH BECOMES

Secondary vesicle	Adult structures
Telencephalon	The cerebrum and the lateral ventricles
Diencephalon	The thalamus, hypothalamus, epithalamus, and third ventricle
Mesencephalon	The midbrain and the cerebral aqueduct
Metencephalon	The pons, the cerebellum, and the upper fourth ventricle
Myelencephalon	The medulla oblongata and the lower fourth ventricle

HOLD ONTO THIS

The secondary vesicle list is the adult brain in order, from front to back: cerebrum, diencephalon, midbrain, pons and cerebellum, medulla. Learn the five vesicles and you have already named the major regions.

The Cerebrum

The cerebrum is the largest region of the brain. Its surface is folded to pack more cortex into the skull, and a thick core of white matter lies beneath.

THE SURFACE AND THE CORE

Structure	What it is
Cerebral hemispheres	The right and left halves of the cerebrum, joined by the corpus callosum, each with five lobes
Cerebral cortex	The outer layer of gray matter, 2 to 4 mm thick, holding billions of neuron cell bodies
Cerebral white matter	The myelinated axons beneath the cortex that connect cortical areas to each other and to lower regions
Gyri	The raised ridges of the folded cortex
Sulci	The shallow grooves between the gyri
Fissures	The deepest grooves, which divide major regions of the brain

Specific named sulci and fissures mark the boundaries on the cerebral surface. Compare them.

THE NAMED SURFACE BOUNDARIES

Structure	What it separates
Central sulcus	The frontal lobe from the parietal lobe; it divides the primary motor cortex in front from the primary somatosensory cortex behind
Lateral sulcus	The frontal and parietal lobes from the temporal lobe
Parieto-occipital sulcus	The parietal lobe from the occipital lobe
Longitudinal fissure	The left and right cerebral hemispheres
Transverse fissure	The cerebrum from the cerebellum

The Cerebral Lobes

Each hemisphere has five lobes. Four are named for the cranial bone above them; the fifth, the insula, lies hidden deep in the lateral sulcus. Compare them.

THE FIVE LOBES, WHERE EACH SITS AND WHAT IT DOES

Lobe	Location	Primary roles
Frontal lobe	The front of the cerebrum	Voluntary movement at the precentral gyrus, motor planning, speech production at Broca's area, judgment, and personality
Parietal lobe	Behind the central sulcus	Somatic sensation at the postcentral gyrus and spatial orientation
Temporal lobe	The lower side of the cerebrum	Hearing, language comprehension at Wernicke's area, and memory
Occipital lobe	The back of the cerebrum	Visual processing
Insula	Deep within the lateral sulcus	Taste, visceral sensation, and emotion

Cerebral White Matter Tracts

The white matter beneath the cortex is sorted into three classes of fiber, grouped by where each tract begins and ends. Compare them.

THE THREE FIBER CLASSES

Fiber class	What it connects	Examples
Association fibers	Gyri within the same hemisphere, allowing communication across one side	The arcuate fasciculus, which links Broca's and Wernicke's areas
Commissural fibers	Matching gyri between the two hemispheres	The corpus callosum, the largest; also the anterior and posterior commissures
Projection fibers	The cortex with lower brain regions and the spinal cord, as ascending or descending tracts	The internal capsule, the major projection pathway beside the basal ganglia

The Basal Ganglia

The basal ganglia, also called the cerebral nuclei, are deep masses of gray matter buried within the cerebral white matter. They help regulate movement.

THE NUCLEI AND WHAT THEY DO

Structure	What it is
Basal ganglia	The cerebral nuclei, deep gray matter masses that control semi-voluntary movement, such as the arm swing of walking
Caudate nucleus	A long, curved nucleus, one of the core components
Putamen	A core component lying lateral to the globus pallidus
Globus pallidus	A core component lying medial to the putamen
Substantia nigra	A midbrain nucleus grouped with the basal ganglia by function
Subthalamic nucleus	A diencephalic nucleus grouped with the basal ganglia by function
Roles	Initiating and regulating movement, suppressing unwanted movement, and modulating muscle tone

Functional Areas of the Cerebral Cortex

The cortex is mapped into specialized areas of three kinds: motor areas that issue commands, sensory areas that receive input, and association areas that interpret and integrate.

THE THREE KINDS OF AREA

Structure	What it is
Motor areas	Issue the commands for voluntary movement; found mainly in the frontal lobe
Sensory areas	Receive and begin to process input from the body's receptors
Association areas	Integrate sensory input with memory and reasoning to plan a response

Each named area has a fixed location and a specific job. Compare them.

THE NAMED FUNCTIONAL AREAS

Area	Location	Role
Primary motor cortex	Precentral gyrus of the frontal lobe	Initiates voluntary movement
Premotor cortex	Anterior to the primary motor cortex	Plans learned, patterned movements
Supplementary motor area	Medial frontal lobe	Coordinates complex bilateral movements
Frontal eye field	Middle frontal gyrus	Controls voluntary eye movement
Broca's area	Inferior frontal gyrus, usually the left	Controls the motor production of speech
Primary somatosensory cortex	Postcentral gyrus of the parietal lobe	Receives conscious touch, pressure, and proprioception
Primary visual cortex	Occipital lobe, at the calcarine sulcus	Receives raw visual input
Primary auditory cortex	Superior temporal gyrus	Receives auditory input
Primary gustatory cortex	Insula and inferior postcentral gyrus	Receives taste input
Primary olfactory cortex	Medial temporal lobe, at the uncus	Receives smell input
Vestibular cortex	Posterior insula and the parietal cortex	Receives input for balance and spatial orientation
Prefrontal cortex	Anterior frontal lobe	Executive function: judgment, planning, and personality
Wernicke's area	The temporal and parietal junction, usually the left	Comprehension of spoken and written language
Somatosensory association area	Posterior to the postcentral gyrus	Interprets texture, shape, and spatial orientation
Visual association area	Surrounding the primary visual cortex	Recognizes faces and objects
Auditory association area	Adjacent to the primary auditory cortex	Interprets speech and music

The Diencephalon

The diencephalon links the cerebral hemispheres to the brainstem and surrounds the third ventricle. It has three parts: the thalamus, the hypothalamus, and the epithalamus.

THE THREE PARTS AND WHAT LIES WITHIN THEM

Structure	What it is
Thalamus	Paired oval masses of gray matter that form about 80 percent of the diencephalon; the main sensory relay station to the cortex
Massa intermedia	The interthalamic adhesion, a bridge joining the right and left thalami
Thalamic nuclei	The individual nuclei of the thalamus, each one relaying a specific sensory or motor modality
Hypothalamus	The region below the thalamus; the master control center for homeostasis
Hypothalamic roles	Controlling the autonomic nervous system, endocrine function through the pituitary, body temperature, hunger and thirst, and sleep-wake rhythms
Epithalamus	The small posterior part of the diencephalon
Pineal gland	The endocrine gland of the epithalamus that secretes melatonin
Habenular nuclei	Nuclei of the epithalamus involved in emotional and visceral responses to smell
Third ventricle	The midline fluid-filled cavity enclosed by the diencephalon

The Cerebellum

The cerebellum is the second largest brain region, sitting in the posterior cranial fossa below the occipital lobes. It coordinates movement, posture, and balance.

THE PARTS OF THE CEREBELLUM

Structure	What it is
Cerebellar hemispheres	The two lateral halves of the cerebellum
Vermis	The narrow midline band connecting the two hemispheres
Cerebellar lobes	Each hemisphere has an anterior, a posterior, and a flocculonodular lobe
Folia	The fine, leaf-like folds of the cerebellar surface
Cerebellar cortex	The outer gray matter of the cerebellum
Arbor vitae	The branching, tree-like pattern of cerebellar white matter
Deep cerebellar nuclei	Gray matter nuclei embedded in the white matter, such as the dentate nucleus

Three pairs of cerebellar peduncles, stalks of white matter, connect the cerebellum to the brainstem. Compare them.

THE THREE PAIRS OF PEDUNCLES

Peduncle	What it connects
Superior cerebellar peduncle	Connects to the midbrain; carries mostly output from the cerebellum
Middle cerebellar peduncle	Connects to the pons; carries input from the pontine nuclei
Inferior cerebellar peduncle	Connects to the medulla; carries input from the spinal cord and vestibular system

The Limbic System

The limbic system is a functional system, not a single structure. Its parts ring the diencephalon and reach into the cerebrum. It sets emotional state and supports memory.

THE LIMBIC STRUCTURES AND THEIR ROLES

Structure	What it is
Limbic system	A functional system spanning the cerebrum and diencephalon that sets emotional state and supports memory
Amygdala	The limbic structure for fear and aggression, linking emotion to memory
Hippocampus	The limbic structure essential for forming new long-term memories
Cingulate gyrus	The limbic gyrus arching above the corpus callosum, involved in expressing emotion
Limbic roles	Establishing emotional states and drives, linking conscious thought to brainstem responses, and supporting long-term memory

Cerebral Arterial Supply: The Circle of Willis

The brain is fed by two arterial systems that join in a ring at the base of the brain, the cerebral arterial circle, also called the Circle of Willis. The ring lets blood cross from one source to another if a vessel narrows or is blocked.

THE TWO CIRCULATIONS AND THE RING THAT JOINS THEM

Structure	What it is
Anterior circulation	The blood supply from the internal carotid arteries; it feeds most of the cerebrum
Posterior circulation	The blood supply from the vertebral and basilar arteries; it feeds the brainstem, cerebellum, and occipital lobe
Cerebral arterial circle	The Circle of Willis, the ring of arteries at the base of the brain that joins the anterior and posterior circulations
Anastomosis	A connection between two arteries that gives blood an alternate route if one path is blocked

The thirteen named arteries of the cerebral arterial supply. Compare what each one feeds.

THE THIRTEEN ARTERIES AND THEIR TERRITORIES

Artery	What it supplies
Anterior cerebral artery	The medial surface of the frontal and parietal lobes
Anterior communicating artery	A short bridge joining the right and left anterior cerebral arteries; it closes the front of the ring
Internal carotid artery	The main feed of the anterior circulation; it enters the skull and divides into the anterior and middle cerebral arteries
Middle cerebral artery	The lateral surface of the cerebrum, including the motor and sensory cortex for the body; the largest branch of the internal carotid
Posterior communicating artery	Joins the internal carotid to the posterior cerebral artery on each side, linking the anterior and posterior circulations

Artery	What it supplies
Posterior cerebral artery	The occipital lobe and the inferior surface of the temporal lobe
Basilar artery	Formed by the union of the two vertebral arteries; it runs up the front of the pons and supplies the brainstem
Superior cerebellar artery	A branch of the basilar artery to the superior surface of the cerebellum
Anterior inferior cerebellar artery (AICA)	A branch of the basilar artery to the anterior and inferior cerebellum
Vertebral arteries	The paired arteries that enter through the foramen magnum and unite to form the basilar artery
Posterior inferior cerebellar artery (PICA)	A branch of the vertebral artery to the inferior cerebellum and the lateral medulla
Anterior spinal artery	A single midline artery formed from both vertebral arteries; it runs down the front of the spinal cord
Posterior spinal arteries	Paired arteries from the vertebral arteries that run down the back of the spinal cord

LOCATION NAMES THE DEFICIT

Because each cerebral lobe and cortical area has its own job, the location of an injury predicts the loss. Damage to the **occipital lobe** takes vision; damage to **Broca's area** takes fluent speech while comprehension stays intact. A clinician can often estimate where a stroke struck from the pattern of symptoms alone.

The **projection fibers** of the **internal capsule** funnel nearly all traffic between the cortex and the body into one narrow band. A small stroke there severs a great deal of wiring at once, which is why such a tiny lesion can weaken an entire side of the body.

The **Circle of Willis** is the brain's insurance policy. Because the ring connects the anterior and posterior circulations, a slow blockage in one vessel can sometimes be bypassed through the communicating arteries. A sudden blockage gives the **anastomosis** no time to open, and the territory beyond it dies.

CHAPTER 4

The Brainstem

THE CENTRAL NERVOUS SYSTEM, THE BRAINSTEM

Covers the three regions of the brainstem, the midbrain, the pons, and the medulla oblongata, the structures within each, and the reticular formation that runs through them. The focus is on the structures and the job each one does.

BY THE END

1. Name the three regions of the brainstem and locate each one.
2. Identify the key structures of the midbrain, the pons, and the medulla oblongata.
3. Describe the reticular formation and the reticular activating system.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The brainstem

drsrennie-stack.github.io/new-build-bio4-solano/brainstem-concept-videos.html

The Brainstem, an Overview

The brainstem connects the cerebrum, the diencephalon, and the cerebellum to the spinal cord. Every ascending and descending tract passes through it, and it holds the nuclei for ten of the twelve cranial nerves.

THE THREE REGIONS AND WHAT RUNS THROUGH THEM

Structure	What it is
Brainstem	The stalk of the brain connecting the higher regions to the spinal cord; the midbrain, pons, and medulla
Midbrain	The most superior region of the brainstem, also called the mesencephalon
Pons	The middle region of the brainstem, a bridge to the cerebellum
Medulla oblongata	The most inferior region of the brainstem, continuous with the spinal cord
Cranial nerve nuclei	The clusters of gray matter in the brainstem that give rise to or receive cranial nerves III through XII
Passing tracts	The brainstem carries every ascending sensory tract and descending motor tract between the brain and the cord

The Midbrain

The midbrain, or mesencephalon, is the short, most superior region of the brainstem. It surrounds the cerebral aqueduct.

THE STRUCTURES OF THE MIDBRAIN

Structure	What it is
Tectum	The roof of the midbrain, made of four rounded bumps called colliculi
Superior colliculi	The upper pair of colliculi; centers for visual reflexes, such as turning the eyes and head toward a sight
Inferior colliculi	The lower pair of colliculi; centers for auditory reflexes, such as turning toward a sound
Cerebral peduncles	The paired stalks on the front of the midbrain that carry descending motor tracts
Substantia nigra	A darkly pigmented midbrain nucleus that helps regulate movement, working with the basal ganglia
Cerebral aqueduct	The narrow channel through the midbrain that carries cerebrospinal fluid from the third to the fourth ventricle
Cranial nerve nuclei	The midbrain holds the nuclei for CN III, the oculomotor nerve, and CN IV, the trochlear nerve

HOLD ONTO THIS

Superior colliculi sit above and serve sight; inferior colliculi sit below and serve sound. Upper for eyes, lower for ears, and both are reflex centers, not the places where you become conscious of the sight or the sound.

The Pons

The pons is the rounded, middle region of the brainstem. Its name means bridge.

THE STRUCTURES OF THE PONS

Structure	What it is
Pons	The middle region of the brainstem; a bridge between the medulla, the midbrain, and the cerebellum
Pontine nuclei	Clusters of gray matter that relay motor information from the cerebral cortex to the cerebellum
Middle cerebellar peduncles	The thick stalks that carry signals from the pontine nuclei into the cerebellum
Fourth ventricle	The cerebrospinal fluid space that lies behind the pons and the upper medulla
Cranial nerve nuclei	The pons holds the nuclei for CN V (trigeminal), CN VI (abducens), CN VII (facial), and CN VIII (vestibulocochlear)

The Medulla Oblongata

The medulla oblongata is the most inferior region of the brainstem. It is continuous with the spinal cord at the foramen magnum and contains centers that keep the body alive.

THE STRUCTURES AND VITAL CENTERS OF THE MEDULLA

Structure	What it is
Medulla oblongata	The most inferior region of the brainstem; it merges into the spinal cord
Pyramids	A pair of ridges on the front of the medulla formed by the descending corticospinal motor tracts
Decussation of the pyramids	The crossover point where most corticospinal fibers pass to the opposite side; it is why one hemisphere controls the opposite side of the body
Cardiac center	The medullary center that adjusts the rate and force of the heartbeat
Respiratory center	The medullary center that sets the basic rhythm of breathing
Vasomotor center	The medullary center that adjusts the diameter of blood vessels, and so blood pressure
Cranial nerve nuclei	The medulla holds nuclei for CN VIII (cochlear), CN IX, CN X, CN XI, and CN XII

The Three Regions Compared

Each region of the brainstem has its own structures and its own set of cranial nerve nuclei. Compare them.

MIDBRAIN, PONS, AND MEDULLA SIDE BY SIDE

Region	Key structures	Cranial nerve nuclei
Midbrain	The tectum, with superior colliculi for visual reflexes and inferior colliculi for auditory reflexes; the cerebral peduncles carrying descending motor tracts; the substantia nigra	III and IV
Pons	A bridge between the medulla, midbrain, and cerebellum; contains the pontine nuclei that relay motor information to the cerebellum	V, VI, VII, and VIII
Medulla oblongata	Contains the pyramids, where the corticospinal tracts decussate; holds the vital centers for heart rate, breathing, and blood vessel diameter	VIII through XII

The Reticular Formation

The reticular formation is not one structure. It is a net of gray matter cell bodies woven through the white matter of the entire brainstem.

THE NETWORK AND WHAT IT DOES

Structure	What it is
Reticular formation	A network of gray matter neurons spread through the core of the brainstem
Reticular activating system	The RAS, the ascending part of the network; it keeps the cortex awake, alert, and attentive
Functions of the RAS	Maintaining consciousness and arousal, holding attention, and filtering out unimportant incoming stimuli so the brain is not overloaded
Descending component	The part that helps set the muscle tone of resting skeletal muscle and coordinates motor activity with autonomic reflexes
Loss of RAS activity	When the ascending system is shut down the result is sleep, and severe damage can cause coma

A SMALL REGION WITH VITAL JOBS

The **medulla** holds the **cardiac**, **respiratory**, and **vasomotor** centers. Because these centers run the heartbeat and breathing, injury to the medulla, unlike injury to a single cortical area, is immediately life-threatening.

The **decussation of the pyramids** in the medulla is the anatomical reason a stroke in the right hemisphere weakens the left side of the body. The motor fibers have already crossed by the time they reach the cord.

The **reticular activating system** explains the link between the brainstem and consciousness. A patient can lose a great deal of cortex and stay awake, but damage to the small ascending reticular network can drop them into a coma. The **substantia nigra** ties the brainstem to movement: its loss is the basis of Parkinson disease.

CHAPTER 5

CNS Meninges and CSF

THE MENINGES AND CEREBROSPINAL FLUID

Covers the cranial and spinal meninges, the dural reflections, the meningeal spaces, where the spinal cord ends and the lumbar puncture, the ventricles, cerebrospinal fluid and its circulation, and the blood-brain barrier.

BY THE END

1. Name the cranial and spinal meninges and the spaces between them.
2. Identify where the spinal cord ends and the layers a lumbar puncture passes.
3. Describe cerebrospinal fluid, where it is found, and where it is made.
4. Trace the circulation of CSF from production to reabsorption.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Meninges and CSF

drsrennie-stack.github.io/new-build-bio4-solano/meninges-concept-videos.html

An Overview

The brain and spinal cord are soft and irreplaceable, so they are protected four ways: by bone, by the meninges, by a cushioning fluid, and by a barrier that screens the blood.

THE FOUR KINDS OF PROTECTION

Structure	What it is
Protection of the CNS	The brain and spinal cord are shielded by bone, by membranes, by fluid, and by a blood barrier
Bony protection	The cranium encloses the brain; the vertebral column encloses the spinal cord
Meninges	The three connective tissue membranes that wrap the brain and spinal cord
Cerebrospinal fluid	The clear fluid that surrounds, cushions, and floats the central nervous system
Blood-brain barrier	The barrier that controls what passes from the blood into the brain tissue

The Cranial Meninges

Three connective tissue membranes wrap the brain. Their Latin names run from tough mother on the outside to tender mother within. Compare them.

THE THREE LAYERS, OUTERMOST TO INNERMOST

Layer	Position	Description
Dura mater	The outermost layer	A tough membrane with two layers: an outer periosteal layer against the skull and an inner meningeal layer; the two are fused except where they part to form the dural venous sinuses
Arachnoid mater	The middle layer	A delicate, lace-like membrane
Pia mater	The innermost layer	A thin membrane, rich in blood vessels, that clings directly to the surface of the brain

ONE DIFFERENCE FROM THE VERTEBRAL CANAL

Structure	What it is
No cranial epidural space	The periosteal dura is attached directly to the skull, so there is no epidural space inside the cranium, unlike the vertebral canal

The Dural Reflections

In a few places the inner meningeal dura folds inward as a partition that anchors the brain and separates its parts. These folds are the dural reflections, also called the dural extensions.

THE FOLDS OF THE MENINGEAL DURA

Structure	What it is
Dural reflections	Inward folds of the meningeal dura that partition the cranial cavity
Falx cerebri	The fold that dips into the longitudinal fissure, separating the right and left cerebral hemispheres
Falx cerebelli	The fold that separates the right and left cerebellar hemispheres
Tentorium cerebelli	The fold that separates the cerebrum above from the cerebellum below

Where the two dural layers part within these folds, they form the dural venous sinuses. The sinuses collect venous blood from the brain, along with the CSF returned by the arachnoid granulations, and drain it toward the heart. Follow the main drainage route.

1. **Superior sagittal sinus** runs along the top edge of the falx cerebri; receives CSF from the arachnoid granulations
2. **Confluence of sinuses** the meeting point of the major sinuses, at the internal occipital protuberance
3. **Transverse sinuses** the paired sinuses running laterally from the confluence
4. **Sigmoid sinuses** the S-shaped sinuses that curve down toward the jugular foramen
5. **Internal jugular veins** carry the blood out of the skull and back toward the heart

The Spinal Meninges

The same three membranes continue down around the spinal cord, with one important difference: the vertebral canal has a true epidural space, and the cranium does not.

THE MEMBRANES AND ANCHORS AROUND THE CORD

Structure	What it is
Spinal dura mater	A single tough layer around the cord, not fused to bone, unlike the two-layered cranial dura
Spinal arachnoid mater	The middle membrane, continuous with the cranial arachnoid
Spinal pia mater	The innermost membrane, clinging to the surface of the cord
Epidural space	The space between the spinal dura and the vertebral bone, filled with fat and veins; the site of an epidural injection
Denticulate ligaments	Tooth-like extensions of pia mater that anchor the cord to the dura along its length
Filum terminale	A thread of pia mater that extends from the end of the cord and anchors it to the coccyx

The Meningeal Spaces

The gaps around and between the meninges are named spaces. Each one matters in the clinic, whether for an injection, a bleed, or a spinal tap. Compare them.

THE THREE NAMED SPACES

Space	Location	Contents
Epidural space	Between the spinal dura and the vertebral bone	Fat and veins; present in the vertebral canal but not the cranium
Subdural space	Between the dura mater and the arachnoid mater	A potential space only, with a thin film of fluid
Subarachnoid space	Between the arachnoid mater and the pia mater	Cerebrospinal fluid; this is where CSF is found around the brain and cord

Where the Spinal Cord Ends

The spinal cord is shorter than the vertebral column. It ends high in the lumbar region, and the subarachnoid space continues below it as a pool of CSF.

THE END OF THE CORD AND WHAT LIES BELOW IT

Structure	What it is
Conus medullaris	The tapered end of the spinal cord, at the level of the L1 to L2 intervertebral disc in the adult
Cauda equina	The hair-like bundle of nerve roots that descends past the conus medullaris within the vertebral canal
Filum terminale	The pia mater thread running from the conus medullaris to anchor on the coccyx
Lumbar cistern	The enlarged subarachnoid space below the conus medullaris, filled with CSF and the floating cauda equina

The Lumbar Puncture

A lumbar puncture, or spinal tap, draws CSF from the lumbar cistern. The needle enters in the midline between the L3 to L4 or L4 to L5 spines, safely below the end of the cord. Follow the structures it passes, superficial to deep.

1. **Skin** the epidermis and dermis of the lower back
2. **Subcutaneous fat** the superficial fascia beneath the skin
3. **Supraspinous ligament** the ligament running along the tips of the spinous processes
4. **Interspinous ligament** the ligament between adjacent spinous processes
5. **Ligamentum flavum** the tough elastic ligament joining the laminae of adjacent vertebrae
6. **Epidural space** the fat-filled and vein-filled space inside the vertebral canal
7. **Dura mater** the tough outer meningeal membrane
8. **Arachnoid mater** the delicate middle meningeal membrane
9. **Subarachnoid space** the needle reaches the lumbar cistern, where CSF is withdrawn

The Ventricles of the Brain

The ventricles are connected, fluid-filled chambers deep within the brain, lined by ependymal cells. Cerebrospinal fluid is made in them. Compare them.

THE VENTRICLES AND HOW THEY CONNECT

Ventricle	Location	Connects to
Lateral ventricles	One in each cerebral hemisphere	The third ventricle, through the interventricular foramina of Monro
Third ventricle	In the diencephalon	The fourth ventricle, through the cerebral aqueduct of Sylvius
Fourth ventricle	Between the brainstem and the cerebellum	The central canal and the subarachnoid space, through three apertures
Central canal	Runs the length of the spinal cord	Continuous above with the fourth ventricle

WHAT LINES THEM AND WHAT MAKES THE FLUID

Structure	What it is
Choroid plexus	Networks of capillaries, wrapped in ependymal cells, that produce CSF; found in the lateral, third, and fourth ventricles
Ependymal cells	The glial cells that line the ventricles and central canal and help circulate CSF

Cerebrospinal Fluid

Cerebrospinal fluid is the clear liquid cushion of the central nervous system. It is made, circulated, and reabsorbed continuously, keeping a steady volume.

WHAT CSF IS, WHERE IT IS, AND WHAT IT DOES

Structure	What it is
Cerebrospinal fluid	CSF, the clear, watery fluid that surrounds and fills the central nervous system
Where CSF is found	In the four ventricles, in the central canal of the spinal cord, and in the subarachnoid space around the whole brain and cord, including the lumbar cistern
Site of production	The choroid plexus within the ventricles
Site of reabsorption	The arachnoid granulations, which return CSF to the venous blood of the dural sinuses
Buoyancy	The fluid floats the brain, reducing its effective weight so it does not crush its own base
Protection	The fluid cushions the CNS against jolts and blows
Chemical stability	The fluid carries nutrients, removes wastes, and keeps a stable environment around the neurons

The Circulation of CSF

Cerebrospinal fluid follows a one-way loop, produced in the ventricles and reabsorbed into the blood. The ventricles below hold choroid plexus, where ependymal cells produce CSF. Follow it in order.

1. **Lateral ventricles** hold choroid plexus, where ependymal cells produce CSF; one ventricle sits in each cerebral hemisphere

2. **Interventricular foramina of Monro** CSF passes through these paired openings into the third ventricle
3. **Third ventricle** in the diencephalon; also holds choroid plexus, adding more CSF
4. **Cerebral aqueduct of Sylvius** CSF flows down this channel through the midbrain
5. **Fourth ventricle** between the pons and the cerebellum; also holds choroid plexus
6. **Apertures of the fourth ventricle** CSF exits through the two lateral apertures of Luschka and the median aperture of Magendie
7. **Subarachnoid space** CSF bathes the whole surface of the brain and spinal cord; some passes into the central canal
8. **Arachnoid granulations** CSF is reabsorbed here into the venous blood of the superior sagittal sinus, completing the loop

HOLD ONTO THIS

CSF is made in the ventricles but reabsorbed outside them, at the arachnoid granulations in the superior sagittal sinus. Anything that blocks the narrow points on that route, the foramina, the aqueduct, or the apertures, backs the fluid up behind it.

The Blood–Brain Barrier

The blood-brain barrier is the structural seal that keeps brain tissue separate from the changing chemistry of the blood. It is built from three components.

THE BARRIER AND ITS THREE COMPONENTS

Structure	What it is
Blood–brain barrier	The BBB, a selective barrier between the blood and the brain tissue
Capillary endothelial cells	The cells lining brain capillaries, joined by tight junctions and lacking the pores of ordinary capillaries
Basement membrane	The supporting layer that wraps the capillary endothelium
Astrocyte end-feet	The foot-like projections of astrocytes that envelop the capillary and help control the barrier

Disorders of the Meninges and CSF

Compare the common disorders by the structure each one affects.

THE DISORDERS AND THE STRUCTURE EACH ONE AFFECTS

Disorder	Structure affected	What it is
Meningitis	The meninges	Inflammation of the meninges, usually from infection; diagnosed by testing CSF drawn at a lumbar puncture
Hydrocephalus	The ventricles	A buildup of CSF that enlarges the ventricles and raises intracranial pressure
Epidural hematoma	The epidural space	Bleeding between the dura mater and the skull
Subdural hematoma	The subdural space	Bleeding between the dura mater and the arachnoid mater
Subarachnoid hemorrhage	The subarachnoid space	Bleeding into the CSF-filled space around the brain

THE CORD ENDS, BUT THE FLUID DOES NOT

The spinal cord stops at the **conus medullaris** near the L1 to L2 disc, yet the subarachnoid space runs lower still, as the **lumbar cistern**. That mismatch is the whole basis of the lumbar puncture: a needle placed between the L3 to L4 or L4 to L5 spines reaches a pool of CSF with only the floating **cauda equina** there to drift aside.

Knowing the layers in order is what makes the tap safe. The needle crosses skin, fat, three ligaments, the **epidural space**, and then the dura and arachnoid before CSF appears. A clinician feels the firm **ligamentum flavum** give way as a landmark.

The named spaces also tell a bleed apart. An **epidural hematoma** sits outside the dura, a **subdural hematoma** just beneath it, and a **subarachnoid hemorrhage** mixes into the CSF itself. Each location is found, and treated, differently.

CHAPTER 6

The Spinal Cord

THE CENTRAL NERVOUS SYSTEM, THE SPINAL CORD

Covers the external anatomy of the cord, its enlargements and terminations, the roots and rootlets that connect it to the spinal nerves, and the internal anatomy of its gray and white matter. The focus is on the structures and the job each one does.

BY THE END

1. Describe the external anatomy of the spinal cord, including its enlargements and terminations.
2. Identify the roots, rootlets, and dorsal root ganglion.
3. Describe the internal anatomy of the gray horns and the white columns and tracts.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The spinal cord

drsrennie-stack.github.io/new-build-bio4-solano/spinal-cord-concept-videos.html

The Spinal Cord, an Overview

The spinal cord runs from the medulla oblongata down the vertebral canal. It carries signals between the brain and the body and is the center for spinal reflexes.

THE CORD, ITS DIMENSIONS, AND WHAT IT DOES

Structure	What it is
Spinal cord	The cord of nervous tissue within the vertebral canal, continuous with the medulla above
Length	About 16 to 18 inches in the adult, running from the medulla to the L1 to L2 level
Diameter	Not uniform; widest in the cervical region and narrowest in the thoracic region
Cord growth	The cord stops lengthening at about 4 to 5 years of age, while the vertebral column keeps growing
Functions	Conducting sensory and motor signals between the brain and the body, and acting as the integration center for spinal reflexes

External Anatomy of the Spinal Cord

Two regions of the cord are visibly wider, and the cord ends well above the end of the vertebral column. Two grooves run its full length.

THE ENLARGEMENTS, THE TERMINATIONS, AND THE GROOVES

Structure	What it is
Cervical enlargement	A widening from C4 to T1 where the nerves of the upper limbs arise
Lumbar enlargement	A widening from T11 to S2 where the nerves of the lower limbs arise
Conus medullaris	The tapered, cone-shaped end of the spinal cord, at the disc between L1 and L2
Filum terminale	A thread of pia mater extending from the conus medullaris that anchors the cord to the coccyx
Cauda equina	The hair-like bundle of nerve roots that descends past the conus medullaris within the vertebral canal
Anterior median fissure	The deep groove down the front of the cord
Posterior median sulcus	The shallow groove down the back of the cord; with the fissure it divides the cord into right and left halves

The Roots and Rootlets

Each side of the cord connects to a spinal nerve through two roots. Tiny rootlets gather into a root, and the two roots join to form the mixed spinal nerve.

FROM ROOTLET TO SPINAL NERVE

Structure	What it is
Rootlets	The small bundles of axons that attach in a row along the cord and merge to form a root
Root	A bundle of axons that connects a spinal nerve to the cord; each spinal nerve has two
Posterior (dorsal) root	The sensory root; it carries sensory axons from receptors in the periphery into the cord
Posterior (dorsal) root ganglion	The swelling on the posterior root that holds the cell bodies of the sensory neurons
Anterior (ventral) root	The motor root; it carries motor axons from the cord out to muscles and glands
Spinal nerve	The mixed nerve formed where the posterior and anterior roots join; it carries both sensory and motor axons
Intervertebral foramen	The gap between adjacent vertebrae where the spinal nerve exits the vertebral canal

HOLD ONTO THIS

Posterior is sensory in, anterior is motor out, and only the posterior root carries a ganglion. If a question puts cell bodies on a root, it is the posterior one.

Internal Anatomy: The Gray Matter

A cross-section of the cord shows a butterfly of gray matter at the core, wrapped in white matter. This is the reverse of the brain, where gray matter is on the outside.

THE GRAY CORE AND ITS BRIDGES

Structure	What it is
Gray matter	The inner, butterfly-shaped region of neuron cell bodies
Central canal	The small cerebrospinal fluid channel at the center of the cord, continuous with the fourth ventricle
Gray commissure	The bridge of gray matter that connects the two sides, surrounding the central canal
Anterior white commissure	The bridge of white matter that connects the two sides of the cord

The gray matter is shaped into projections called horns. Compare what each horn holds.

THE GRAY HORNS AND THEIR CONTENTS

Horn	Contents
Posterior (dorsal) horn	Interneurons and the incoming axons of sensory neurons
Lateral horn	Autonomic motor nuclei; present only in the thoracic and lumbar segments
Anterior (ventral) horn	Somatic motor nuclei that drive skeletal muscle contraction

Internal Anatomy: The White Matter

The white matter of the cord is sorted into columns, and each column holds tracts. A tract is a bundle of axons that share a common origin and destination.

THE COLUMNS AND THE TRACTS THEY HOLD

Structure	What it is
White matter	The outer region of the cord, made of myelinated axons
Columns	The anterior, posterior, and lateral white columns, the major divisions of the cord's white matter
Tracts	Bundles of axons within the columns that share a common origin and destination
Ascending tracts	Sensory tracts that conduct impulses up toward the brain; also called afferent tracts
Descending tracts	Motor tracts that conduct impulses down away from the brain; also called efferent tracts

A cluster of cell bodies and a bundle of axons each have a different name in the CNS and the PNS. Compare the four terms.

THE FOUR TERMS BY LOCATION

Location	Cluster of cell bodies	Bundle of axons
Central nervous system	Nucleus	Tract
Peripheral nervous system	Ganglion	Nerve

Distinguishing the Cord Segments

A cross-section taken at different levels of the cord does not look the same. The proportions of gray and white matter, the shape, and the size all shift from top to bottom. Compare them.

TELLING THE CORD SEGMENTS APART IN CROSS-SECTION

Segment	Distinguishing features
Cervical	Large diameter with a large amount of white matter; oval in shape
Thoracic	Small diameter with a small amount of gray matter; has a small lateral gray horn
Lumbar	Almost circular; less white matter than the cervical segments
Sacral	Relatively small, with more gray matter and little white matter
Coccygeal	The smallest; resembles the lower sacral segments

HOLD ONTO THIS

White matter thins as you go down the cord, because fewer tracts are still running, while gray matter grows where a limb needs serving. That is why a cervical section is mostly white and a sacral section is mostly gray.

WHERE THE CORD ENDS, AND WHAT THAT ALLOWS

The spinal cord ends at the **conus medullaris** near the L1 to L2 disc, but the vertebral column runs much lower. Below the conus, only the loose **cauda equina** hangs in the canal. That gap is the safe target for a lumbar puncture: a needle can draw cerebrospinal fluid there without risking the cord itself.

The **posterior root ganglion** is the home address of every sensory neuron cell body for that segment. The varicella virus hides in these ganglia for decades; when it reactivates as shingles, the rash follows the exact strip of skin served by that one ganglion.

A spinal cord injury interrupts the **ascending** and **descending tracts**. Because the tracts carry every signal between the brain and the body below, function is lost below the level of the injury, while everything above it is spared.

CHAPTER 7

The Peripheral Nervous System

THE PERIPHERAL NERVOUS SYSTEM

Covers the structure of a nerve, the spinal nerves and their roots, the rami, the nerve plexuses, and the classification of sensory receptors. The cranial nerves are covered on their own sheet. The focus is on the structures and the job each one does.

BY THE END

1. Describe the structure of a nerve and name its connective tissue wrappings.
2. Identify the spinal nerves, their roots, and their rami.
3. Classify the sensory receptors by stimulus, location, and structure.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The peripheral nervous system

drsrennie-stack.github.io/new-build-bio4-solano/pns-concept-videos.html

The Peripheral Nervous System, an Overview

The peripheral nervous system is everything outside the brain and spinal cord. It carries input from the body's receptors in to the CNS, and carries the response back out to the effectors.

THE COMPONENTS OF THE SYSTEM

Structure	What it is
Peripheral nervous system	All neural structures outside the brain and spinal cord
Function	To carry sensory input from receptors to the CNS, and to carry motor output from the CNS to the effectors
Sensory (afferent) receptors	Structures that detect changes in the internal and external environment
Peripheral nerves	Bundles of axons that connect the CNS to the body
Ganglia	Clusters of neuron cell bodies located outside the CNS
Motor (efferent) endings	The terminals where motor neurons meet the muscles and glands
Cranial nerves	The twelve pairs of nerves that arise from the brain, covered on their own sheet
Spinal nerves	The thirty one pairs of nerves that arise from the spinal cord

The Structure of a Nerve

A nerve is a cordlike organ: many axons bundled together and wrapped in three layers of connective tissue.

THE NERVE AND ITS WRAPPINGS

Structure	What it is
Nerve	A cordlike organ of the PNS; a bundle of peripheral axons, some myelinated and some not, enclosed by connective tissue
Axon	A single nerve fiber; many axons make up a nerve
Fascicle	A bundle of axons grouped together
Endoneurium	The innermost wrapping, around each individual axon
Perineurium	The middle wrapping, the thicker layer around each fascicle
Epineurium	The outermost wrapping, around the entire nerve

HOLD ONTO THIS

The three wrappings nest from the inside out, and each one names the level it surrounds: endoneurium around one axon, perineurium around one fascicle, epineurium around the whole nerve.

The Spinal Nerves

Thirty one pairs of spinal nerves connect the cord to the body. Every one is a mixed nerve, carrying both sensory and motor axons.

THE SPINAL NERVE, ITS ROOTS, AND ITS EXIT

Structure	What it is
Spinal nerves	The thirty one pairs of nerves that arise from the spinal cord
Mixed nerves	Every spinal nerve carries both sensory and motor axons
Posterior (dorsal) root	The sensory root that brings axons into the cord
Anterior (ventral) root	The motor root that carries axons out of the cord
Intervertebral foramen	The opening between adjacent vertebrae where a spinal nerve exits the vertebral canal

The thirty one pairs are grouped by the region of the cord they arise from. Compare the count in each region.

THE SPINAL NERVES BY REGION

Region	Number of pairs
Cervical	Eight pairs
Thoracic	Twelve pairs
Lumbar	Five pairs
Sacral	Five pairs
Coccygeal	One pair

HOLD ONTO THIS

The posterior root brings signals in and the anterior root carries them out. Posterior is sensory, anterior is motor, and the two join just past the cord to make one mixed spinal nerve.

The Rami of a Spinal Nerve

Just past the intervertebral foramen, each spinal nerve splits into branches called rami. Each ramus serves a different territory. Compare them.

THE RAMI AND WHAT EACH ONE SUPPLIES

Ramus or branch	What it supplies
Posterior (dorsal) ramus	The deep muscles and the skin of the posterior surface of the trunk
Anterior (ventral) ramus	The muscles and skin of the limbs and the lateral and anterior trunk; these rami form the plexuses
Meningeal branch	Re-enters the vertebral canal to supply the vertebrae, the ligaments, the blood vessels of the cord, and the meninges
Rami communicantes	Small branches off the anterior ramus that carry autonomic fibers to and from the sympathetic trunk

The Nerve Plexuses

The anterior rami of most spinal nerves do not run straight to their targets. They first join into networks called plexuses. The thoracic rami are the exception.

THE PLEXUSES AND THE THORACIC EXCEPTION

Structure	What it is
Nerve plexus	A network formed by the anterior rami of several spinal nerves
Cervical plexus	Formed from C1 to C4; serves the neck and shoulders, and gives off the phrenic nerve to the diaphragm
Brachial plexus	Formed from C5 to T1; serves the shoulder and the upper limb
Lumbar plexus	Formed from L1 to L4; serves the lower abdominal wall and part of the lower limb
Sacral plexus	Formed from L4 to S4; serves the buttock, the perineum, and most of the lower limb
Intercostal nerves	The anterior rami of T2 to T12, which do not form a plexus but run straight to the thoracic wall

The plexuses are covered in more detail, including the region affected by damage, on the Nerve Plexuses sheet.

Classifying Sensory Receptors

Sensory receptors are classified in three ways: by the type of stimulus they detect, by where they are located, and by their structure.

By stimulus type

WHAT KIND OF CHANGE THE RECEPTOR ANSWERS TO

Structure	What it is
Mechanoreceptors	Respond to mechanical force such as touch, pressure, vibration, and stretch
Thermoreceptors	Respond to changes in temperature
Photoreceptors	Respond to light; found in the retina of the eye
Chemoreceptors	Respond to chemicals in solution, such as smell and taste molecules
Nociceptors	Respond to potentially damaging stimuli and produce the sensation of pain

By location

WHERE THE RECEPTOR SITS

Structure	What it is
Exteroceptors	Detect stimuli at or near the body surface, such as touch, temperature, and the special senses
Interoceptors	Also called visceroceptors; detect stimuli inside the body, in the organs and blood vessels
Proprioceptors	Detect the position and movement of the body, in muscles, tendons, and joints

By structure

HOW THE NERVE ENDING IS BUILT

Structure	What it is
Free nerve endings	Nonencapsulated receptors; bare dendrite tips found in most tissues
Encapsulated receptors	Nerve endings wrapped in a connective tissue capsule

The Sensory Receptors

The named sensory receptors, with where each one is found and what it detects. Compare them.

THE NAMED RECEPTORS, THEIR LOCATION, AND THEIR STIMULUS

Receptor	Location	What it detects
Free nerve endings	Most body tissues, dense in connective tissue and epithelium	Pain, temperature, and crude touch
Tactile (Merkel) discs	The deepest layer of the epidermis	Light, discriminative touch
Hair follicle receptors	Wrapped around hair follicles	Movement of the hair
Tactile (Meissner) corpuscles	The dermal papillae of hairless skin	Fine, discriminative touch
Lamellar (Pacinian) corpuscles	The deep dermis and the hypodermis	Deep pressure and vibration
Bulbous (Ruffini) corpuscles	The deep dermis and joint capsules	Continuous, steady pressure
Muscle spindles	Within skeletal muscles	The stretch of a muscle
Tendon organs	Within tendons	The tension a muscle places on its tendon
Joint kinesthetic receptors	The capsules of synovial joints	Joint position and movement

FOLLOWING THE NERVE TO THE DEFICIT

Because a nerve is wrapped in connective tissue, it has some protection and some ability to heal. The **epineurium** and **perineurium** hold a cut nerve's shape, which is what lets a surgeon line up the fascicles and repair it.

The rami explain referred patterns. The **anterior rami** carry most limb signals, so a problem in the anterior rami or the plexus they form affects the limbs, while a **posterior ramus** problem stays in the muscles and skin of the back.

Receptors are not spread evenly. The fingertips and lips are crowded with tactile corpuscles and Merkel discs, which is why those areas feel fine detail so well, while the skin of the back, with far fewer, senses touch only crudely.

CHAPTER 8

The Cranial Nerves

THE PERIPHERAL NERVOUS SYSTEM, THE CRANIAL NERVES

Covers the twelve pairs of cranial nerves: the number and name of each, its classification as sensory, motor, or mixed, its function, the skull foramen it passes through, and its nuclei. The focus is on the structures and the job each one does.

BY THE END

1. Name the twelve cranial nerves in order and give the number of each.
2. Classify each cranial nerve as sensory, motor, or mixed.
3. Identify the function, skull foramen, and nuclei of each cranial nerve.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The cranial nerves

drsrennie-stack.github.io/new-build-bio4-solano/cranial-nerves-concept-videos.html

The Cranial Nerves, an Overview

Twelve pairs of cranial nerves arise from the brain. They are numbered I to XII from front to back by where they attach. The first two attach to the cerebrum and diencephalon; the rest attach to the brainstem.

THE VOCABULARY OF THE CRANIAL NERVES

Structure	What it is
Cranial nerves	Twelve pairs of nerves that arise from the brain and brainstem, part of the peripheral nervous system
Numbering	The nerves are numbered with Roman numerals I to XII, from anterior to posterior
Olfactory (I) and optic (II)	The only cranial nerves that attach above the brainstem, to the cerebrum and the diencephalon
Brainstem nerves	Cranial nerves III through XII all attach to the brainstem, where their nuclei sit
Nucleus	The cluster of cell bodies in the brain that a cranial nerve arises from or projects to
Foramen	The opening in the skull that a cranial nerve passes through to reach its target

Classifying the Cranial Nerves

Every cranial nerve is classed by the kind of signal it carries: sensory only, motor only, or both. A nerve carrying both is called mixed.

THE THREE CLASSES AND THE KINDS OF SIGNAL

Structure	What it is
Sensory nerves	Carry only sensory signals into the brain; CN I, CN II, and CN VIII
Motor nerves	Carry mainly motor signals out to muscles; CN III, CN IV, CN VI, CN XI, and CN XII
Mixed nerves	Carry both sensory and motor signals; CN V, CN VII, CN IX, and CN X
Special sensory	The senses of smell, vision, taste, hearing, and balance
Somatic motor	Voluntary control of skeletal muscle
Visceral motor	Parasympathetic control of smooth muscle and glands; carried by CN III, VII, IX, and X

A memory aid for the order of the names is the sentence "On Old Olympus' Towering Tops A Finn And German Viewed Some Hops." For the sensory, motor, or both pattern, "Some Say Marry Money But My Brother Says Big Brains Matter Most."

The Twelve Cranial Nerves

The twelve nerves in order, with the classification, the main function, and the skull foramen each one passes through. Compare them.

NUMBER, NAME, CLASS, FUNCTION, AND FORAMEN

Number	Name	Class	Primary function	Skull foramen
CN I	Olfactory	Sensory	Smell	Cribriform plate of the ethmoid
CN II	Optic	Sensory	Vision	Optic canal of the sphenoid
CN III	Oculomotor	Motor	Most eye movements, raising the eyelid, and constricting the pupil	Superior orbital fissure
CN IV	Trochlear	Motor	Eye movement by the superior oblique muscle; the smallest cranial nerve	Superior orbital fissure
CN V	Trigeminal	Mixed	Sensation from the face and the muscles of chewing; the largest cranial nerve, with three branches	Superior orbital fissure (V1), foramen rotundum (V2), and foramen ovale (V3)
CN VI	Abducens	Motor	Eye movement by the lateral rectus muscle	Superior orbital fissure
CN VII	Facial	Mixed	Facial expression, taste from the front of the tongue, and tear and saliva production	Internal acoustic meatus, then the stylomastoid foramen
CN VIII	Vestibulocochlear	Sensory	Hearing and balance	Internal acoustic meatus
CN IX	Glossopharyngeal	Mixed	Taste from the back of the tongue, swallowing, and saliva from the parotid gland	Jugular foramen
CN X	Vagus	Mixed	Autonomic control of the thoracic and abdominal organs, plus voice and swallowing	Jugular foramen
CN XI	Accessory	Motor	Movement of the sternocleidomastoid and trapezius muscles	Jugular foramen
CN XII	Hypoglossal	Motor	Movement of the tongue	Hypoglossal canal

HOLD ONTO THIS

Three foramina carry more than one nerve. The superior orbital fissure takes III, IV, V1, and VI, the internal acoustic meatus takes VII and VIII, and the jugular foramen takes IX, X, and XI. Learn the openings and the groupings come with them.

Cranial Nerve Nuclei

Each cranial nerve connects to one or more nuclei, the clusters of cell bodies in the brain that the nerve arises from or carries signals to. Most sit in the brainstem.

WHERE EACH NERVE BEGINS OR ENDS IN THE BRAIN

Cranial nerve	Nuclei or origin
CN I, Olfactory	The olfactory epithelium projects to the olfactory bulb, then along the olfactory tract to the cortex
CN II, Optic	The retina projects through the optic nerve and the optic chiasm to the lateral geniculate nucleus

Cranial nerve	Nuclei or origin
CN III, Oculomotor	The oculomotor nucleus and the Edinger-Westphal nucleus of the midbrain
CN IV, Trochlear	The trochlear nucleus of the midbrain
CN V, Trigeminal	The trigeminal ganglion and the trigeminal nuclei of the pons
CN VI, Abducens	The abducens nucleus of the pons
CN VII, Facial	The facial motor nucleus and the solitary nucleus of the pons
CN VIII, Vestibulocochlear	The cochlear and vestibular nuclei at the pons and medulla
CN IX, Glossopharyngeal	The solitary nucleus, the nucleus ambiguus, and the inferior salivatory nucleus of the medulla
CN X, Vagus	The dorsal motor nucleus, the nucleus ambiguus, and the solitary nucleus of the medulla
CN XI, Accessory	The spinal accessory nucleus of the medulla and the upper spinal cord, C1 to C5
CN XII, Hypoglossal	The hypoglossal nucleus of the medulla

TWELVE NERVES, TWELVE EXAMS

Because each cranial nerve has one job and one path, testing the nerves one by one is a fast way to localize a problem in the brainstem. A clinician who finds **CN VI** weak knows to look at the pons, where its nucleus sits.

The foramina group nerves together. The **jugular foramen** carries CN IX, X, and XI side by side, so a tumor or fracture there can knock out all three at once, producing a recognizable cluster of swallowing, voice, and shoulder signs.

Two of the most common nerve problems are cranial. **Bell's palsy** is a sudden weakness of CN VII that drops one side of the face. **Trigeminal neuralgia** is a stabbing facial pain along the branches of CN V. The anatomy of the nerve predicts exactly where each one shows up.

CHAPTER 9

The Nerve Plexuses

THE PERIPHERAL NERVOUS SYSTEM, THE NERVE PLEXUSES

A brief reference for the nerve plexuses. A plexus is a network where the anterior rami of spinal nerves regroup before traveling to the body. Covers the principal plexuses at a general level: where each one forms and what region it serves, so you can predict the area of the body affected if a plexus is damaged.

BY THE END

1. Define a nerve plexus and explain why the anterior rami regroup.
2. Name the principal nerve plexuses and the spinal nerves that form each.
3. Predict the region of the body affected if a given plexus is damaged.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The nerve plexuses

drsrennie-stack.github.io/new-build-bio4-solano/pns-concept-videos.html

What a Nerve Plexus Is

Most spinal nerves do not run straight to their targets. The anterior rami of neighboring spinal nerves first weave together into a network called a plexus, then sort their axons into the nerves that leave it. The thoracic region is the exception.

THE PLEXUS AND WHY IT EXISTS

Structure	What it is
Nerve plexus	A network where the anterior rami of several spinal nerves join and regroup before their axons travel on to the body
Anterior rami	The branches of the spinal nerves that form the plexuses and supply the limbs and the front of the body
Why a plexus forms	Regrouping lets each body region receive axons from several spinal nerves, so damage to one spinal nerve does not erase all function in that area
Intercostal nerves	The exception; the anterior rami of T2 to T12 do not form a plexus but run directly to the muscles and skin of the thoracic wall
Principal plexuses	The cervical, brachial, lumbar, sacral, and coccygeal plexuses, each named for the region it serves

The Principal Nerve Plexuses

Each plexus is formed by a known set of spinal nerves and serves a known region of the body. Compare them.

THE FIVE PLEXUSES, WHAT FORMS EACH, AND THE REGION IT SERVES

Plexus	Formed by	Region served
Cervical plexus	The anterior rami of C1 to C4, with a contribution from C5	The skin and muscles of the neck and the upper shoulders; it also gives rise to the phrenic nerve, the motor supply to the diaphragm
Brachial plexus	The anterior rami of C5 to C8 and T1	The shoulder and the entire upper limb
Lumbar plexus	The anterior rami of L1 to L4	The anterolateral abdominal wall, the external genitals, and part of the lower limb
Sacral plexus	The anterior rami of L4 to L5 and S1 to S4	The buttocks, the perineum, and most of the lower limb
Coccygeal plexus	The anterior rami of S4 to S5 and the coccygeal nerves	A small area of skin over the coccyx

HOLD ONTO THIS

The plexuses run head to tail in the same order as the cord regions that build them: cervical for the neck, brachial for the upper limb, lumbar for the abdominal wall and thigh, sacral for the buttock and lower limb, coccygeal for the skin over the coccyx.

Predicting the Effect of Plexus Damage

Because each plexus serves one region, the location of a plexus injury predicts the region of the body that loses movement and sensation. This is the level of detail to know for this course.

INJURY SITE AND THE REGION AFFECTED

Plexus damaged	Region affected
Cervical plexus	Weakness or numbness of the neck and shoulder; if the phrenic nerve is involved, the diaphragm on that side is weakened and breathing is affected
Brachial plexus	Weakness and sensory loss across the shoulder and the whole upper limb on that side
Lumbar plexus	Weakness and sensory loss of the lower abdominal wall and the front and inner surfaces of the thigh
Sacral plexus	Weakness and sensory loss of the buttock, the back of the thigh, and the leg and foot
Coccygeal plexus	Numbness of the small patch of skin over the coccyx

WHY THE NETWORK MATTERS

The plexus design is a safety feature. Because each limb region draws axons from several spinal nerves, a single damaged spinal nerve root weakens an area but rarely paralyzes it completely. The overlap covers the gap.

Damage to the **plexus** itself is different. A plexus injury sits downstream of that regrouping, so it can take out a whole region at once. The level of the injury predicts the deficit: a brachial plexus injury affects the upper limb, a lumbar or sacral injury the lower limb.

The **phrenic nerve** is the one to remember. It leaves the cervical plexus and is the sole motor supply to the diaphragm. The teaching phrase "C3, 4, 5 keep the diaphragm alive" is a reminder that a high cervical injury can stop breathing.

CHAPTER 10

The Autonomic Nervous System

THE AUTONOMIC NERVOUS SYSTEM

Covers the difference between the somatic and autonomic systems, the two-neuron motor pathway, the named autonomic ganglia, the sympathetic and parasympathetic divisions, and the autonomic and enteric plexuses. The focus is on the structures and the job each one does.

BY THE END

1. Contrast the somatic and autonomic motor systems.
2. Describe the two-neuron motor pathway and name the autonomic ganglia.
3. Compare the sympathetic and parasympathetic divisions and the plexuses.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The autonomic nervous system

drsrennie-stack.github.io/new-build-bio4-solano/ans-concept-videos.html

The Autonomic Nervous System, an Overview

The autonomic nervous system is the involuntary motor division. It quietly regulates the organs, adjusting them without conscious effort or awareness.

WHAT THE SYSTEM IS AND WHAT IT CONTROLS

Structure	What it is
Autonomic nervous system	The involuntary division of the motor system; it controls cardiac muscle, smooth muscle, and glands
Autonomic effectors	The targets of the ANS: cardiac muscle, smooth muscle, and glands
Autonomic sensory neurons	Neurons that carry input from receptors in the viscera and blood vessels to the CNS
Autonomic motor neurons	Neurons that carry commands out to the visceral effectors, either exciting or inhibiting them
Two divisions	The sympathetic and the parasympathetic divisions, which usually act in opposite directions on the same organ
Involuntary control	Most autonomic activity cannot be consciously altered, although emotion and pain can influence it

Somatic Versus Autonomic

The somatic and autonomic systems are both motor, but they differ in what they control, how many neurons they use, and what effect they have. Compare them.

THE TWO MOTOR SYSTEMS SIDE BY SIDE

Feature	Somatic nervous system	Autonomic nervous system
Effector	Skeletal muscle	Cardiac muscle, smooth muscle, and glands
Control	Voluntary	Involuntary
Motor neurons in the pathway	One, from the CNS straight to the muscle	Two, with a synapse in a ganglion
Effect on the effector	Always excitation; it causes contraction	Either excitation or inhibition
Awareness	Normally conscious	Normally not conscious

The Two-Neuron Motor Pathway

Where the somatic system reaches its muscle with a single neuron, the autonomic system always uses two neurons in a row, with a synapse in a ganglion between them.

THE TWO NEURONS AND THE GANGLION BETWEEN THEM

Structure	What it is
Preganglionic neuron	The first neuron; its cell body sits in the brain or spinal cord, and its myelinated axon runs out to an autonomic ganglion
Autonomic ganglion	The cluster of cell bodies where the two neurons synapse
Postganglionic neuron	The second neuron; its cell body sits in the ganglion, and its unmyelinated axon runs from the ganglion to the visceral effector
The pathway	The preganglionic neuron synapses in the ganglion with the postganglionic neuron, which then reaches the effector

HOLD ONTO THIS

The preganglionic axon is myelinated and the postganglionic axon is not. The synapse in the ganglion is the dividing line, and it is the one feature the somatic system does not have at all.

The Autonomic Ganglia

The autonomic ganglia fall into two groups by division. The sympathetic ganglia sit close to the spinal cord; the parasympathetic ganglia sit close to or within the organ.

Sympathetic ganglia

THE GANGLIA NEAR THE VERTEBRAL COLUMN

Structure	What it is
Sympathetic trunk ganglia	Also called paravertebral ganglia; a vertical chain on each side of the vertebral column, from the base of the skull to the coccyx
Prevertebral ganglia	Also called collateral ganglia; ganglia anterior to the vertebral column, near the large abdominal arteries
The five prevertebral ganglia	The celiac, superior mesenteric, inferior mesenteric, aorticorenal, and renal ganglia

Parasympathetic ganglia

THE GANGLIA NEAR THE TARGET ORGAN

Structure	What it is
Terminal ganglia	The parasympathetic ganglia, located near or within the wall of the target organ
The four named terminal ganglia of the head	The ciliary, pterygopalatine, submandibular, and otic ganglia

The Two Divisions

The two divisions are told apart first by where they leave the CNS, and then by the work they do.

The sympathetic division

FIGHT OR FLIGHT

Structure	What it is
Sympathetic division	The fight or flight division; also called the thoracolumbar division
Origin	Preganglionic cell bodies in the lateral gray horns of the T1 to L2 spinal segments
Role	It takes over during exercise, excitement, emergency, and embarrassment
Sample effects	Dilates the pupils, speeds the heart, opens the airways, and slows digestion

The parasympathetic division

REST AND DIGEST

Structure	What it is
Parasympathetic division	The rest and digest division; also called the craniosacral division
Origin	Preganglionic cell bodies in the nuclei of four cranial nerves (III, VII, IX, and X) and in the S2 to S4 spinal segments
Role	It conserves energy and runs routine housekeeping: digestion, defecation, and urination
Sample effects	Constricts the pupils, slows the heart, constricts the airways, and stimulates digestion

Comparing the Two Divisions

Side by side, the two divisions differ in their origin in the CNS, the location of their ganglia, and their overall job. Compare them.

SYMPATHETIC AGAINST PARASYMPATHETIC

Feature	Sympathetic	Parasympathetic
Other name	Thoracolumbar division	Craniosacral division
Origin in the CNS	The T1 to L2 spinal segments	The brainstem and the S2 to S4 spinal segments
Location of ganglia	Close to the spinal cord	Close to or within the effector organ
Overall role	Fight or flight; prepares the body for action	Rest and digest; conserves energy

HOLD ONTO THIS

The two other names are the whole map of the origins. Thoracolumbar means the fibers leave from T1 to L2, and craniosacral means they leave from the brainstem and S2 to S4. The ganglia then follow: sympathetic near the cord, parasympathetic at the organ.

The Autonomic and Enteric Plexuses

Autonomic axons and ganglia travel together in networks called plexuses. A separate set of networks, the enteric plexuses, runs the wall of the gut.

Autonomic plexuses

THE NAMED NETWORKS OF THE TRUNK

Structure	What it is
Autonomic plexus	A network of autonomic axons and ganglia in the thorax, abdomen, and pelvis
Cardiac plexus	Supplies the heart
Pulmonary plexus	Supplies the bronchial tree
Celiac (solar) plexus	The largest; it surrounds the celiac trunk and supplies most of the abdominal organs
Superior and inferior mesenteric plexuses	Supply the small and large intestine
Hypogastric plexus	Supplies the pelvic organs
Renal plexus	Supplies the kidneys and ureters

Enteric plexuses

THE NETWORKS IN THE WALL OF THE GUT

Structure	What it is
Enteric plexuses	The neural networks within the wall of the GI tract, the structural core of the enteric nervous system
Myenteric plexus	Lies between the longitudinal and circular muscle layers; it controls the movement of the gut
Submucosal plexus	Lies between the circular muscle layer and the muscularis mucosae; it controls the secretions of the gut

Tracing an Autonomic Pathway

Every autonomic motor command travels the same four-step route from the central nervous system to the organ.

1. **Preganglionic neuron** the cell body in the brain or cord sends its axon out toward a ganglion
2. **Autonomic ganglion** the preganglionic axon synapses here with the postganglionic neuron
3. **Postganglionic neuron** its axon carries the signal the rest of the way out
4. **Effector tissue** the cardiac muscle, smooth muscle, or gland receives the command

TWO DIVISIONS, OPPOSITE EFFECTS

Because the two divisions usually pull an organ in opposite directions, the body can fine-tune an organ by easing off one division rather than only by pushing the other. Heart rate, for example, rises when **parasympathetic** input is withdrawn, even before the sympathetic division speeds things up.

The **vagus nerve**, CN X, carries most of the parasympathetic supply to the chest and abdomen. Its long reach is why a single nerve can slow the heart, ease breathing, and drive digestion all at once.

The **enteric nervous system** is independent enough to be called the brain of the gut. Smooth muscle in the GI tract keeps contracting in a rhythm and glands keep secreting even when the link to the CNS is cut, which is why a denervated bowel can still move food along.

PART B

Pre-work.

5 packets, in the order they are due. Work them in sequence: read the Part A chapter, organize it into mini visuals, draw it from memory, apply it with your notes closed, then retrieve later in the week by redrawing from memory. Context before content. This is not a night-before task.

THESE PAGES PRINT LANDSCAPE. TURN THE PACKET SIDEWAYS.

Pre-work · Nervous Tissue and Neuronal Integration

Functional Organization and Nervous Tissue, and Gross Anatomy and Neuronal Integration. Read both Part A chapters first. This packet does not repeat them. Every other packet in Module 5 is built on the cell and the vocabulary in these two chapters, so the time you spend here is repaid four more times.

Module 5 · 1 of 5 | Bring this to class

How to work this pre-work

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.

2. **Organize what you read.**
In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.

3. **Draw it.**
In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.

4. **Apply it.**
Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.

5. **Come back to it later in the week**
, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

PANEL 1 OF 3

Organize it

Turn the two tissue chapters into mini visuals. Eight chunks. The grids carry the naming pairs, the hubs carry the

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second color and mark what you missed.

structure, and the one chain is the reflex arc.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

WORKED EXAMPLE

A patient cuts a nerve in the forearm and, over many months, sensation and movement return. Another patient has a tract severed in the spinal cord and recovers nothing. Both injuries cut axons. Explain the difference from structure.

Answer. The forearm nerve is myelinated by Schwann cells, which leave a neurolemma. The cord tract is myelinated by oligodendrocytes, which do not.

Reasoning. Start with the cell that builds the sheath, not with the axon. In the PNS a Schwann cell wraps one segment of one axon, and its outer layer, the neurolemma, survives the injury as a hollow tube. That tube is a guide, so the regrowing axon has a path back to its target. In the CNS one oligodendrocyte reaches up to fifteen axons and leaves no neurolemma behind, so a regrowing axon has nothing to follow. The location of the injury also matters, because an axon cut off from its cell body loses its supply of proteins and dies beyond the cut. So the answer is not about the axon at all, it is about which glial cell was doing the wrapping.

Set A · Name it

1. The gaps in the myelin sheath along an axon are the ________________________.
2. A cluster of neuron cell bodies located outside the CNS is a ________________________.
3. The glial cell that lines the ventricles and helps circulate cerebrospinal fluid is the ________________________.
4. The cone-shaped region where the cell body joins the axon is the ________________________.
5. Name the five components of a reflex arc, in order.

____

____

1 HUB

The nervous system and its divisions

Nervous system at the hub. Spoke first to the CNS with the brain and spinal cord, then to the PNS with nerves, ganglia, and sensory receptors. Off the motor side of the PNS hang somatic, autonomic, and enteric, and off autonomic hang sympathetic and parasympathetic. Write the effector each motor division reaches at the end of its spoke.

2 HUB

The neuron

One cell body in the middle. Spoke out to dendrites, axon hillock, axon with axoplasm and axolemma, and axon terminals. Inside the cell body mark the nucleus, the Nissl bodies, the neurofibrils, and the microtubules. Draw an arrow along the whole cell showing which way an impulse travels, because the direction is what names the parts.

DRAWING 1

A multipolar neuron, fully labelled

BRAIN DUMP FIRST

Two minutes. Every part of the cell, and the direction a signal moves through it.

Draw one large multipolar neuron. Give it many branched dendrites, one cell body with the nucleus and Nissl bodies inside, a clear axon hillock, and one long axon ending in branched terminals with synaptic end bulbs. Wrap the axon in a myelin sheath with nodes of Ranvier between the segments. Now enlarge one end bulb in a bubble and draw the synaptic vesicles, the cleft, and the next cell's membrane.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5–4 A multipolar neuron, fully labelled** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can put your finger on any part of your drawing and say whether a signal at that point is travelling toward the cell body or away from it.

Set B · Predict it

1. The immune system strips myelin from axons in the CNS. Predict the effect on impulse speed and name the cell that was lost.
2. An axon is cut halfway along its length. Predict what happens to the portion beyond the cut and explain why, using the cell body.
3. A neuronal pool is wired for divergence. Predict what one incoming signal does to the number of neurons responding.
4. A stroke damages a descending pathway on the right side of the brain. Predict which side of the body is weakened and name the reason.

Set C · Tell it apart

1. Distinguish a tract from a nerve, and a nucleus from a ganglion, by what each is made of and where it sits.
2. Explain why a tumor of nervous tissue almost always arises from glia rather than from neurons.
3. Distinguish a monosynaptic from a polysynaptic reflex by the wiring in the integration center, and give an example of each.

3 GRID

Classifying neurons, twice

Neuron type down the side: multipolar, bipolar, unipolar. Number of processes, where it is found, and which functional class usually wears that shape across the top. Then add two rows at the bottom for Purkinje and pyramidal cells and write where each one lives.

4 GRID

The six neuroglia

Glial cell down the side, all six. CNS or PNS, and the job it does across the top. Add a third column: what goes wrong if this cell fails. Four rows are CNS and two are PNS, and keeping them grouped is half the learning.

5 SPLIT

PNS myelination against CNS myelination

One split. Which cell builds the sheath, how many axons one cell can reach, neurolemma present or absent, and whether a cut axon can regrow. This is the pair that answers the regeneration question, so make the neurolemma row impossible to miss.

DRAWING 2

A myelinated axon in the PNS and in the CNS, side by side

BRAIN DUMP FIRST

One minute. Which cell wraps each one, and what is left behind on the outside.

Draw two axons beside each other. On the left, wrap the PNS axon with Schwann cells, one per segment, and draw the neurolemma as the outer layer of each Schwann cell. On the right, draw one oligodendrocyte with several arms, each arm wrapping a segment of a different CNS axon, and leave the outer surface bare. Mark the nodes of Ranvier on both and draw the impulse jumping node to node.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5-5 A myelinated axon in the PNS and in the CNS, side by side** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain from your two drawings why a cut nerve in the forearm can regrow and a cut tract in the cord usually cannot.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

6 SPLIT

Gray matter against white matter

One split. What each is made of, and where each sits. Then hang two small sketches off the branches: a spinal cord cross-section with gray inside, and a brain section with gray outside. The swap between the two organs is the whole point of the visual.

7 GRID

Tract, nerve, nucleus, ganglion

A true two by two. Bundle of axons and cluster of cell bodies down the side, CNS and PNS across the top. Write the term in each of the four cells, then add one example of each. If you can fill this grid blank you have four terms for the price of one.

8 CHAIN

The reflex arc, five components

Five boxes with arrows, receptor through to effector. Write beside each box where that component physically sits, and mark the one box that is inside the CNS. Then note beside the integration center whether the arc is monosynaptic or polysynaptic.

DRAWING 3

Two reflexes drawn as arcs

BRAIN DUMP FIRST

Two minutes. All five components in each, and the difference in the middle.

Draw a spinal cord cross-section twice. On the first, draw the patellar reflex: the muscle spindle receptor, the sensory neuron entering the posterior root, one synapse in the cord, the motor neuron leaving the anterior root, and the quadriceps as the effector. On the second, draw the withdrawal reflex with an interneuron added between the sensory and motor neurons. Label all five components on both and mark the posterior root ganglion.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5-6 Two reflexes drawn as arcs** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can count the synapses on each of your drawings and name the reflex type from the count alone.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Mini visual sheet

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

The nervous system and its divisions

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

2

The neuron

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

3

Classifying neurons, twice

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

4

The six neuroglia

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

5

PNS myelination against CNS myelination

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

6

Gray matter against white matter

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

7

Tract, nerve, nucleus, ganglion

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

8

The reflex arc, five components

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

Drawing pages

The drawings from Panel 2, with room to draw them. Brain dump first with everything closed, then draw, then correct yourself in a second color.

DRAWING 1

A multipolar neuron, fully labelled

DRAWING 2

A myelinated axon in the PNS and in the CNS, side by side

DRAWING 3

Two reflexes drawn as arcs

Pre-work · The Brain and Brainstem

The Brain, and the Brainstem. Read both Part A chapters first. Almost every question on this material is a location question: name the region, then name what it does. Learn the map before you learn the list, because the map is what makes the list stick.

Module 5 · 2 of 5

Bring this to class

How to work this pre-work

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
   Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.

2. **Organize what you read.**
   In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.

3. **Draw it.**
   In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.

4. **Apply it.**
   Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.

5. **Come back to it later in the week**
   , not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

PANEL 1 OF 3

Organize it

Turn the brain and brainstem chapters into mini visuals. Eight chunks. The hubs carry the regions, the grids carry the functional detail, and the chain is the

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second color and mark what you missed.

development story that explains why the regions sit where they do.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 CHAIN

Neural tube to adult brain

One chain with arrows. Neural tube, then the three primary vesicles, then the five secondary vesicles, then the adult structure each one becomes. Write the ventricle that forms alongside each secondary vesicle, because the fluid spaces develop with the tissue around them.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

WORKED EXAMPLE

A patient speaks in short, effortful phrases and struggles to get words out, but follows every spoken instruction correctly and can read. Name the area damaged, the gyrus it sits in, and the lobe.

Answer. Broca's area, in the inferior frontal gyrus of the frontal lobe, usually on the left.

Reasoning. Split language into its two halves before you name anything. Comprehension of spoken and written language belongs to Wernicke's area at the temporal and parietal junction, and comprehension is intact here, so that area is ruled out. What is lost is the motor production of speech, and motor areas of the cortex sit in the frontal lobe, which narrows it at once. Broca's area is the one frontal area named for speech production, and it lies in the inferior frontal gyrus, usually on the left. Notice that the arcuate fasciculus links these two areas, so a lesion in that association bundle would produce a different pattern again, and the pattern of what is spared is what tells the three apart.

Set A · Name it

1. The sulcus separating the primary motor cortex from the primary somatosensory cortex is the ______________________.
2. The commissural tract joining the two cerebral hemispheres is the ______________________.
3. The branching, tree-like white matter of the cerebellum is the ______________________.
4. The ridges on the front of the medulla formed by the descending motor tracts are the ______________________.
5. Name the five secondary brain vesicles and one adult structure formed by each.

____

____

2 HUB

The five cerebral lobes

One lateral view of a hemisphere at the hub. Spoke to frontal, parietal, temporal, occipital, and insula, and write the primary role of each at the end of its spoke. Draw the central sulcus, the lateral sulcus, and the parieto-occipital sulcus as the boundaries, and mark the insula as hidden inside the lateral sulcus.

3 GRID

The functional areas of the cortex

Area down the side, motor areas first, then sensory, then association. Lobe, exact gyrus or sulcus, and role across the top. Sixteen rows is a lot, so build it in three blocks and check each block by covering the location column.

DRAWING 1

The lateral surface of the cerebrum, with the functional map on it

BRAIN DUMP FIRST

Two minutes. Five lobes, the sulci that bound them, and the named areas in position.

Draw one hemisphere from the side, gyri and sulci included. Draw the central sulcus, the lateral sulcus, and the parieto-occipital sulcus, then outline and label all five lobes, showing the insula peeled open inside the lateral sulcus. Now mark the primary motor cortex on the precentral gyrus, the primary somatosensory cortex on the postcentral gyrus, the primary visual, auditory, and gustatory areas, Broca's area, Wernicke's area, and the prefrontal cortex.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5–7 The lateral surface of the cerebrum, with the functional map on it** so you can find it later. Correct it in a second color.

Set B · Predict it

1. A small stroke destroys part of the internal capsule. Predict how much of the body is affected and explain why so small a lesion does so much.

2. The substantia nigra degenerates. Predict the kind of deficit this produces and name the system it belongs to by function.

3. An injury damages the ascending reticular network of the brainstem. Predict the effect on the patient and say why cortical damage does not do the same thing.

4. The posterior cerebral artery is occluded. Predict which lobe loses its supply and what function is lost.

Set C · Tell it apart

1. Distinguish the superior from the inferior colliculi by what each one serves and by what they do not do.

2. Explain why an injury to the medulla is immediately life-threatening while an injury to one cortical area is not.

3. A slow narrowing of one internal carotid artery causes fewer symptoms than a sudden blockage of the same vessel. Explain, using the ring at the base of the brain.

4 LADDER

The three classes of cerebral white matter fiber

Three rungs, ordered by how far each fiber reaches: within one hemisphere, between the two hemispheres, and out of the cerebrum entirely. Write the class name and one named example beside each rung. The bottom rung carries the internal capsule, and that is the one the clinical questions are built on.

5 HUB

The deep brain in coronal section

The third ventricle at the hub. Spoke out to the thalamus with the massa intermedia, the hypothalamus below it, and the epithalamus with the pineal gland behind. Then move laterally to the caudate nucleus, putamen, and globus pallidus, and draw the internal capsule threading between them. Write the job of each structure beside it.

YOU KNOW IT WHEN

You can be given any point on your own drawing and say which lobe it is in and what is lost if that spot is damaged.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

6 HUB

The cerebellum

The cerebellum at the hub. Spoke to the two hemispheres, the vermis between them, the folia of the surface, the cerebellar cortex, the arbor vitae, and the deep nuclei. Then draw the three pairs of peduncles reaching to the midbrain, the pons, and the medulla, and mark which pair carries output rather than input.

7 GRID

Midbrain, pons, and medulla compared

Region down the side, superior to inferior. Key structures, cranial nerve nuclei, and one thing that region alone does across the top. Star the medulla row, because the vital centers are the reason an injury there behaves differently from an injury anywhere else in the brain.

DRAWING 2

The brainstem, front and back

BRAIN DUMP FIRST

Two minutes. All three regions and every surface feature on each.

Draw the brainstem twice, once from the front and once from behind with the cerebellum removed. On the front view show the cerebral peduncles of the midbrain, the bulge of the pons, and the pyramids of the medulla with the decussation at the bottom. On the back view show the four colliculi of the tectum, the fourth ventricle, and the three pairs of cerebellar peduncles. Then write the cranial nerve nuclei held by each region beside it.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5–8 The brainstem, front and back** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can name the region a cranial nerve nucleus sits in from the nerve number alone, and point to that region on your drawing.

8 SPLIT

Anterior circulation against posterior circulation

One split. Internal carotid on one branch, vertebral and basilar on the other. Name the arteries that come off each and write the territory each one feeds. Then draw the three communicating arteries that join the two branches into a closed ring and label the ring.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 3

The brain in midsagittal section

BRAIN DUMP FIRST

Two minutes. Every region on the cut midline, cerebrum down to medulla.

Draw the brain sliced down the midline. Lay in the corpus callosum as the arch across the middle and the cingulate gyrus curving above it. Below the corpus callosum draw the thalamus, the hypothalamus, and the pineal gland of the epithalamus around the third ventricle. Continue down through the midbrain, pons, and medulla, and add the cerebellum behind them with the arbor vitae drawn inside it.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5-9 The brain in midsagittal section** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can trace the four major parts of the brain in order on your own drawing and name the boundary between each pair.

Mini visual sheet

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

Neural tube to adult brain

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

2

The five cerebral lobes

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

3

The functional areas of the cortex

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

4

The three classes of cerebral white matter fiber

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

5

The deep brain in coronal section

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

6

The cerebellum

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

7

Midbrain, pons, and medulla compared

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

8

Anterior circulation against posterior circulation

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

Drawing pages

The drawings from Panel 2, with room to draw them. Brain dump first with everything closed, then draw, then correct yourself in a second color.

DRAWING 1

The lateral surface of the cerebrum, with the functional map on it

DRAWING 2

The brainstem, front and back

DRAWING 3

The brain in midsagittal section

Pre-work · Meninges, CSF, and the Spinal Cord

The Meninges and Cerebrospinal Fluid, and the Spinal Cord. Read both Part A chapters first. These two chapters are layers and routes rather than lists. Work them with your finger tracing the order, and the naming comes with it.

Module 5 · 3 of 5 | Bring this to class

How to work this pre-work

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
   Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.

2. **Organize what you read.**
   In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.

3. **Draw it.**
   In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.

4. **Apply it.**
   Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.

5. **Come back to it later in the week**
   , not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

PANEL 1 OF 3

Organize it

Turn the meninges and cord chapters into mini visuals. Eight chunks. The ladders carry the layers, the chain carries the fluid, and the hubs carry the

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second color and mark what you missed.

two structures you will be asked to draw from memory.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 **LADDER**

The meninges, outside in

Rungs from the skull down to the brain surface: periosteal dura, meningeal dura, arachnoid, pia. Write the space between each pair of rungs on the line between them, and mark the one space that holds cerebrospinal fluid. Then note beside the top rung why the cranium has no epidural space and the vertebral canal does.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

WORKED EXAMPLE

A clinician places a needle in the midline between the L4 and L5 spinous processes and withdraws clear fluid. Name the pool the needle reached and explain why the spinal cord is not injured.

Answer. The lumbar cistern, the enlarged subarachnoid space below the conus medullaris.

Reasoning. The answer comes from a mismatch in growth, not from the needle. The spinal cord stops lengthening at about four to five years of age while the vertebral column keeps growing, so in the adult the cord ends at the conus medullaris near the L1 to L2 disc, well above the entry point. The meninges do not stop there: the subarachnoid space runs on down as the lumbar cistern, so there is still cerebrospinal fluid to draw. What occupies that canal below the conus is the cauda equina, a loose bundle of nerve roots that floats in the fluid and drifts aside from a blunt needle. An entry above L2 would be a different question entirely, because the cord itself would be in the way, and that is why the level is part of the answer.

Set A · Name it

1. The meningeal layer that clings directly to the surface of the brain is the ____________________.
2. The dural fold separating the cerebrum from the cerebellum is the ____________________.
3. The network of capillaries wrapped in ependymal cells that produces CSF is the ____________________.
4. The tapered inferior end of the spinal cord is the ____________________.
5. Name the three components of the blood-brain barrier.

____

2 GRID

The three meningeal spaces

Space down the side. Location, contents, and one clinical use or bleed across the top. Add a fourth column: present in the cranium, yes or no. The subdural row is the one people get wrong, so write potential space in it and underline it.

3 HUB

The dural reflections and the sinuses they carry

A brain in the cranial cavity at the hub. Draw the falx cerebri dipping into the longitudinal fissure, the tentorium cerebelli between cerebrum and cerebellum, and the falx cerebelli below it. Then draw the superior sagittal sinus in the top edge of the falx cerebri and follow it back to the confluence, the transverse sinuses, the sigmoid sinuses, and the internal jugular veins.

DRAWING 1

The meninges over the top of the brain, in coronal section

BRAIN DUMP FIRST

Two minutes. Every layer from bone to brain, and every space between them.

Draw a wedge of skull with the brain beneath it, cut across the midline. Lay in the periosteal dura against the bone and the meningeal dura beneath it, and show the two layers parting at the midline to enclose the superior sagittal sinus. Draw the falx cerebri dipping down into the longitudinal fissure between the hemispheres. Add the arachnoid mater with arachnoid granulations poking up into the sinus, the subarachnoid space filled with CSF, and the pia mater following every gyrus and sulcus.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5-1 The meninges over the top of the brain, in coronal section** so you can find it later. Correct it in a second color.

Set B · Predict it

1. The cerebral aqueduct is narrowed by a tumor. Predict which ventricles enlarge and which do not, and name the condition.
2. The arachnoid granulations are scarred and stop working. Predict what happens to CSF volume and pressure, and say why production does not simply stop.
3. A cross-section of cord shows a small lateral horn. Predict which region of the cord it came from and give your reason.
4. A cord injury is sustained at the T6 level. Predict which functions are lost and which are spared, using the tracts.

Set C · Tell it apart

1. Distinguish an epidural hematoma from a subdural hematoma from a subarachnoid hemorrhage by the space each one fills.
2. Explain why the vertebral canal has a true epidural space and the cranium does not, using the layers of the dura.
3. Compare a cervical cord section with a sacral cord section by the proportion of gray to white matter, and explain the reason for the difference.

4 GRID

The four cerebrospinal fluid spaces

Ventricle down the side, lateral through to the central canal. Location, what it connects to, and the name of the opening it connects through across the top. Add a fourth column: does it hold choroid plexus. Three rows say yes and one says no.

5 CHAIN

The circulation of CSF, made to reabsorbed

Eight boxes with arrows, lateral ventricles through to the arachnoid granulations. Mark every narrow point on the route with a circle: the interventricular foramina, the cerebral aqueduct, and the apertures of the fourth ventricle. Those three circles are where a blockage produces hydrocephalus.

YOU KNOW IT WHEN

You can name every layer your needle would cross going from the scalp to the cortex, and say which of them a cranial epidural bleed sits outside of.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

6 LADDER

The layers a lumbar puncture crosses

Nine rungs, skin at the top and the subarachnoid space at the bottom. Write the three ligaments in the correct order in the middle, and mark the rung where the clinician feels the resistance give way. Beside the ladder write the vertebral level the needle enters and the level the cord ends at.

7 HUB

The spinal cord in cross section

The central canal at the hub. Draw the gray butterfly around it with the posterior, lateral, and anterior horns, and write what each horn holds. Then draw the white matter around the gray as anterior, posterior, and lateral columns. Add the rootlets, both roots, the posterior root ganglion, and the spinal nerve leaving on one side only.

DRAWING 2

The ventricles and the whole path of the CSF

BRAIN DUMP FIRST

Two minutes. Four spaces, three narrow points, and where the fluid leaves the system.

Draw the brain in outline from the side and put the ventricles inside it: a lateral ventricle curving through the hemisphere, the third ventricle in the diencephalon, the cerebral aqueduct through the midbrain, and the fourth ventricle between the brainstem and the cerebellum, continuing into the central canal. Mark the choroid plexus in each ventricle that has it. Then arrow the fluid out through the apertures into the subarachnoid space, over the surface, and into the superior sagittal sinus at the arachnoid granulations.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5-2 The ventricles and the whole path of the CSF** so you can find it later. Correct it in a second color.

❽ **GRID**

Telling the cord segments apart

Segment down the side, cervical through coccygeal. Diameter, amount of white matter, amount of gray matter, and shape across the top. Then add one column for the lateral horn, present or absent, because only two segments have it.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

YOU KNOW IT WHEN

You can point to any one narrow spot on your drawing and say which ventricles would enlarge if it were blocked.

DRAWING 3

The spinal cord, whole and in cross section

BRAIN DUMP FIRST

Two minutes. The outside of the cord, where it ends, and what a slice through it shows.

Draw the whole cord from the medulla down, inside the vertebral canal. Mark the cervical enlargement and the lumbar enlargement, taper the cord into the conus medullaris at the L1 to L2 disc, run the filum terminale down to the coccyx, and hang the cauda equina around it. Then draw one cross-section beside it: anterior median fissure, posterior median sulcus, gray horns, white columns, central canal, both roots, and the posterior root ganglion.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5-3 The spinal cord, whole and in cross section** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain from your own drawing why a needle placed between L4 and L5 draws fluid without touching the spinal cord.

Mini visual sheet

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

The meninges, outside in

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

2

The three meningeal spaces

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

3

The dural reflections and the sinuses they carry

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

4

The four cerebrospinal fluid spaces

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

5

The circulation of CSF, made to reabsorbed

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

6

The layers a lumbar puncture crosses

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

7

The spinal cord in cross section

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

8

Telling the cord segments apart

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

Drawing pages

The drawings from Panel 2, with room to draw them. Brain dump first with everything closed, then draw, then correct yourself in a second color.

DRAWING 1

The meninges over the top of the brain, in coronal section

DRAWING 2

The ventricles and the whole path of the CSF

DRAWING 3

The spinal cord, whole and in cross section

Pre-work · The Peripheral and Autonomic Nervous Systems

The Peripheral Nervous System, the Cranial Nerves, the Nerve Plexuses, and the Autonomic Nervous System. Read all four Part A chapters first. This is the largest packet in the module because it is four chapters, and the twelve cranial nerves are learned by mnemonic, so build yours in Panel 2 and use it all week.

Module 5 · 4 of 5 | Bring this to class

How to work this pre-work

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
   Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.

2. **Organize what you read.**
   In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.

3. **Draw it.**
   In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.

4. **Apply it.**
   Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.

5. **Come back to it later in the week**
   , not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

PANEL 1 OF 3

Organize it

Turn the four peripheral chapters into mini visuals. Eight chunks. The chains carry the pathways, the grids carry the two lists you have to memorize, and the

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second color and mark what you missed.

split is the pair that the whole autonomic chapter is built on.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 **LADDER**

A nerve, from axon to whole nerve

Three rungs, single axon at the bottom, fascicle in the middle, whole nerve at the top. Write the connective tissue wrapping beside each rung, and note which axons inside are myelinated and which are not. The names give away the level each one wraps, so write that out beside the ladder.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

WORKED EXAMPLE

A tumor grows at the jugular foramen on the right. The patient has difficulty swallowing, a hoarse voice, and weakness turning the head and shrugging the right shoulder. Tongue movement and facial expression are normal. Name the nerves involved.

Answer. Cranial nerves IX, X, and XI.

Reasoning. Work from the opening, because a foramen is a place where several nerves are squeezed together and injured as a set. The jugular foramen carries the glossopharyngeal, vagus, and accessory nerves side by side, and the three deficits described match those three jobs exactly: swallowing and taste from the back of the tongue for IX, voice and swallowing for X, and the sternocleidomastoid and trapezius for XI. Now check what is spared, because that is what rules out the neighbours. Tongue movement is normal, so the hypoglossal nerve is intact, and it leaves by a different opening, the hypoglossal canal. Facial expression is normal, so the facial nerve is intact, and it travels the internal acoustic meatus instead. Learn the foramina and the clusters of signs come with them.

Set A · Name it

1. The connective tissue wrapping around a single axon within a nerve is the ________________.
2. The nerve arising from the cervical plexus that supplies the diaphragm is the ________________.
3. The receptor of the deep dermis that detects deep pressure and vibration is the ________________.
4. The parasympathetic ganglia, located near or within the wall of the target organ, are the ________________.
5. Write a mnemonic of your own for the twelve cranial nerve names in order, and a second one for the sensory, motor, or mixed pattern.

2 CHAIN

From rootlet to ramus

One chain with arrows. Rootlets, posterior root with its ganglion and anterior root, then the mixed spinal nerve at the intervertebral foramen, then the split into posterior ramus, anterior ramus, meningeal branch, and rami communicantes. Write what each of the four branches supplies at the end of the chain.

3 GRID

The twelve cranial nerves

Number and name down the side, I through XII. Class as sensory, motor, or mixed, primary function, and skull foramen across the top. Then circle the three foramina that carry more than one nerve and bracket the nerves that share each one. Write both memory sentences down the margin before you leave this frame.

DRAWING 1

A spinal nerve, from the cord to its rami

BRAIN DUMP FIRST

Two minutes. Both roots, the ganglion, the mixed nerve, and every branch off it.

Draw a cord cross-section inside a vertebra. Bring rootlets off the posterior and anterior surfaces, gather them into the two roots, put the posterior root ganglion on the sensory root, and join the roots into the mixed spinal nerve at the intervertebral foramen. Then split the nerve into the posterior ramus going to the back, the anterior ramus going around the trunk, the meningeal branch turning back into the canal, and the rami communicantes running to a sympathetic trunk ganglion. Color sensory and motor axons differently the whole way.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5–10 A spinal nerve, from the cord to its rami** so you can find it later. Correct it in a second color.

Set B · Predict it

1. A patient loses movement of the sternocleidomastoid and trapezius on one side. Name the nerve and predict the other two nerves at risk with it.
2. An injury damages the brachial plexus on the left. Predict the region affected and explain why damage to a single spinal nerve root would be less complete.
3. A patient is startled by a loud noise. Predict what the sympathetic division does to the pupils, the heart, the airways, and digestion.
4. The vagus nerve is cut low in the neck. Predict which organs lose parasympathetic supply and explain how one nerve reaches so many.

Set C · Tell it apart

1. Distinguish the somatic from the autonomic motor pathway by the number of neurons, the effect on the effector, and awareness.
2. Compare where the sympathetic and parasympathetic ganglia sit, and explain how that difference changes the length of each axon.

4 **HUB**

The sensory receptors, drawn where they live

A block of skin at the hub, epidermis down to hypodermis, with muscle and tendon beside it. Place each named receptor at its true depth: Merkel discs in the deepest epidermis, Meissner corpuscles in the dermal papillae, free nerve endings and hair follicle receptors through the dermis, Pacinian corpuscles deep, Ruffini corpuscles deep, then muscle spindles, tendon organs, and joint receptors. Write what each one detects beside it.

5 **GRID**

The five plexuses

Plexus down the side, cervical through coccygeal. Formed by which anterior rami, region served, and the deficit if it is damaged across the top. Then write the one exception underneath: the anterior rami that form no plexus at all, and where they go instead.

YOU KNOW IT WHEN

You can start at a receptor in the skin of the back and follow one continuous line into the cord on your own drawing, naming every structure you cross.

3. Distinguish the myenteric from the submucosal plexus by position in the gut wall and by what each one controls.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

6 SPLIT

Somatic against autonomic

One split. Effector, voluntary or involuntary, number of motor neurons in the pathway, effect on the effector, and whether you are aware of it. Five rows on each branch. The row that names the number of neurons is the one everything else follows from, so put it first.

7 GRID

Sympathetic against parasympathetic

Feature down the side, sympathetic and parasympathetic across the top. Other name, origin in the CNS, location of the ganglia, overall role, and four sample effects on named organs. The two other names are a map of the origins, so write them at the top and check the origin row against them.

DRAWING 2

The twelve cranial nerves on the base of the brain, and your own mnemonic

BRAIN DUMP FIRST

Two minutes. All twelve, in order, attached where they belong.

Draw the inferior surface of the brain: frontal lobes at the front, temporal lobes to the sides, the three brainstem regions down the middle, and the cerebellum behind. Attach all twelve nerves in order and number them, with I and II reaching the cerebrum and diencephalon and III through XII on the brainstem. Then write out the two memory sentences from Part A, one for the names and one for the sensory, motor, or mixed pattern. Now build your own for each, using words that mean something to you, and write both under your drawing.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5-11 The twelve cranial nerves on the base of the brain, and your own mnemonic** so you can

8 CHAIN

The two-neuron autonomic pathway

Four boxes with arrows: preganglionic neuron, autonomic ganglion, postganglionic neuron, effector tissue. Mark which axon is myelinated and which is not. Then draw the chain twice, once with the ganglion near the cord and once with it at the organ, and label which division each version belongs to.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can recite your own mnemonic and expand every word into the correct nerve name and class, with the drawing covered.

DRAWING 3

The two autonomic divisions, side by side

BRAIN DUMP FIRST

Two minutes. Where each one leaves the CNS, where its ganglia sit, and how far each neuron runs.

Draw the brain and spinal cord twice, side by side. On the sympathetic side mark the T1 to L2 lateral horns as the origin, draw the sympathetic trunk as a chain of ganglia beside the vertebral column, add the prevertebral ganglia near the abdominal arteries, and run a short preganglionic and a long postganglionic axon out to an organ. On the parasympathetic side mark the nuclei of CN III, VII, IX, and X and the S2 to S4 segments, and run a long preganglionic axon to a terminal ganglion at the organ with a short postganglionic axon beyond it.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5-12 The two autonomic divisions, side by side** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can look at either drawing and say which axon is long, which is short, and why the ganglion position decides it.

Mini visual sheet

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

A nerve, from axon to whole nerve

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

2

From rootlet to ramus

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

3

The twelve cranial nerves

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

4

The sensory receptors, drawn where they live

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

5

The five plexuses

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

6

Somatic against autonomic

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

7

Sympathetic against parasympathetic

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

8

The two-neuron autonomic pathway

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

Drawing pages

The drawings from Panel 2, with room to draw them. Brain dump first with everything closed, then draw, then correct yourself in a second color.

DRAWING 1

A spinal nerve, from the cord to its rami

DRAWING 2

The twelve cranial nerves on the base of the brain, and your own mnemonic

DRAWING 3

The two autonomic divisions, side by side

Pre-work · Module 5 Integration

All ten Part A chapters. This is a synthesis sheet, not a new chapter. Nothing on it is material you have not already read, and that is the point: everything here crosses at least two chapters, so it is the sheet that shows you what you organized separately and never joined together. Work it with the notes closed and open them only to correct.

Module 5 · 5 of 5 | Bring this to class

How to work this pre-work

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.

2. **Organize what you read.**
In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.

3. **Draw it.**
In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.

4. **Apply it.**
Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.

5. **Come back to it later in the week**
, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

PANEL 1 OF 3

Organize it

Rebuild Module 5 as eight visuals that each cross more than one chapter. Nothing here belongs to a single sheet, so if a chunk feels easy you are

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second color and mark what you missed.

probably answering it from one chapter only.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 **HUB**

The whole nervous system on one page

The CNS at the hub, brain and cord drawn in outline. Spoke out to the twelve cranial nerves, the thirty one spinal nerves, the five plexuses, the ganglia, the sensory receptors, and the three kinds of effector. Write the count beside every spoke that has one. If a count is missing you have found a gap.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

WORKED EXAMPLE

A patient suddenly develops weakness of the right arm and right leg and speaks in short, effortful phrases, though comprehension is intact. Name the hemisphere affected, the artery most likely blocked, and explain both.

Answer. The left hemisphere, and the left middle cerebral artery.

Reasoning. Start with the side, because the crossing settles it. Most corticospinal fibers decussate at the pyramids of the medulla, so a hemisphere controls the opposite half of the body, and right-sided weakness points to the left hemisphere. Now use the second deficit to locate the lesion within that hemisphere: speech production is lost while comprehension is intact, which is Broca's area in the inferior frontal gyrus, usually on the left. One artery supplies both the lateral motor cortex for the body and that frontal speech area, and it is the middle cerebral artery, the largest branch of the internal carotid. The anterior cerebral artery is ruled out because it feeds the medial surface, and the posterior cerebral artery is ruled out because it feeds the occipital lobe and vision is unaffected. The Circle of Willis could not rescue the territory here because a sudden blockage gives the anastomosis no time to open.

Set A · Name it

1. The only root of a spinal nerve that carries a ganglion is the ______________________.
2. The glial cell that forms myelin in the CNS is the ______________________.
3. The crossover of motor fibers on the front of the medulla is the ______________________.
4. The enlarged subarachnoid space below the end of the spinal cord is the ______________________.
5. Name the four ways the central nervous system is protected.

____

❷ CHAIN

One stimulus, all the way through

One long chain with arrows. Receptor in the skin, sensory neuron, posterior root and its ganglion, posterior horn, ascending tract, decussation, thalamus, somatosensory cortex, then motor cortex, internal capsule, cerebral peduncle, pyramids, anterior horn, anterior root, spinal nerve, skeletal muscle. Mark the two places the pathway crosses the midline.

❸ SPLIT

CNS against PNS, every term that changes

One split. Cluster of cell bodies, bundle of axons, myelinating cell, neurolemma present or absent, regeneration possible or not, and whether gray matter is inside or outside. Six rows on each branch, and every row is a pair you have already met on a different sheet.

DRAWING 1

The brain, from memory, nothing open

BRAIN DUMP FIRST

Three minutes. Every region of the brain and brainstem, and the fluid spaces inside them.

Draw the brain in midsagittal section from memory alone. Put in the cerebrum with the corpus callosum and the cingulate gyrus, the thalamus, hypothalamus, and pineal gland around the third ventricle, the midbrain with its colliculi and aqueduct, the pons, the medulla, and the cerebellum with the arbor vitae and the fourth ventricle in front of it. Then add the falx cerebri and the tentorium cerebelli in position and mark the superior sagittal sinus.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5–13 The brain, from memory, nothing open** so you can find it later. Correct it in a second color.

Set B · Predict it

1. A high cervical cord injury occurs at the C3 level. Predict the effect on breathing and name the nerve and plexus responsible.

2. A lesion sits in the lateral gray horn of the T4 segment. Predict which kind of neuron is lost and which division of the nervous system is affected.

3. Myelin is stripped from axons in the white columns of the cord. Predict the effect on signals travelling to and from the brain, and say why the same injury in a peripheral nerve behaves differently.

4. A blockage forms at the median aperture of the fourth ventricle. Predict where CSF backs up and where the fluid still reaches normally.

Set C · Tell it apart

1. Trace a signal from a receptor in the fingertip to the cortex and back out to the muscle, naming every structure it passes through.

2. Explain why a lesion of the anterior horn and a lesion of the posterior root ganglion produce opposite deficits.

3. Compare the location of a somatic motor neuron cell body with the locations of the two

❹ **GRID**

Where every cell body sits

Neuron down the side: somatic sensory, somatic motor, interneuron, sympathetic preganglionic, sympathetic postganglionic, parasympathetic preganglionic, parasympathetic postganglionic. Location of the cell body, and CNS or PNS across the top. This grid is the single hardest thing in the module to hold, and it is worth the frame.

❺ **LADDER**

From the scalp inward to the neuron

Rungs from the outside in: bone, periosteal dura, meningeal dura, arachnoid, subarachnoid space with CSF, pia, cerebral cortex as gray matter, white matter beneath. Then add the blood-brain barrier as a note beside the cortex rungs with its three components. Four kinds of protection are on this ladder, so mark each one.

YOU KNOW IT WHEN

You can complete this drawing without opening anything, and every correction you then make in a second color is a named gap rather than a vague feeling.

autonomic motor neuron cell bodies, and explain what the ganglion adds.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

❻ GRID

Cranial nerve nuclei by brainstem region

Region down the side, midbrain, pons, and medulla. Which cranial nerve nuclei sit there, and one other structure of that region across the top. Then add a top row for the two nerves that attach above the brainstem, and name what they attach to instead.

❼ GRID

Six lesions and the deficits they name

Lesion site down the side: the internal capsule, the medulla, Broca's area, the jugular foramen, the phrenic nerve, and a CNS tract stripped of myelin. Structure damaged, deficit produced, and the chapter it came from across the top. Fill it from memory before you check it.

DRAWING 2

A spinal cord segment with everything attached to it

BRAIN DUMP FIRST

Two minutes. Cord, coverings, roots, nerve, and the autonomic chain beside it.

Draw one cord segment in cross-section inside a vertebra. Add the gray horns and white columns, then wrap it in pia, arachnoid with the subarachnoid space, and dura, with the epidural space outside that. Bring out the rootlets, both roots, the posterior root ganglion, and the mixed spinal nerve, then split it into all four branches. Finally put a sympathetic trunk ganglion beside the vertebral body and connect it with the rami communicantes.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5-14 A spinal cord segment with everything attached to it** so you can find it later. Correct it in a second color.

8 CHAIN

CSF, choroid plexus back to the blood

Eight boxes with arrows, from the lateral ventricles to the arachnoid granulations. This is the second time you have drawn this route, so do it with the packet closed and time yourself. Under thirty seconds means you have it.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

YOU KNOW IT WHEN

You can name every structure a needle would pass through from the skin of the back to the central canal, using only this drawing.

DRAWING 3

The somatic and autonomic motor pathways, side by side

BRAIN DUMP FIRST

Two minutes. Both routes drawn from the CNS to the effector, with every difference visible.

Draw the cord twice. On the left run the somatic pathway: one motor neuron from the anterior horn, out the anterior root, along the spinal nerve, to a skeletal muscle. On the right run the autonomic pathway: a preganglionic neuron from the lateral horn out to a ganglion, a synapse, then a postganglionic neuron to smooth muscle, cardiac muscle, or a gland. Mark which axons are myelinated, and label the effect each pathway can have on its effector.

Room to draw this is on the drawing pages at the end of this pre-work.

In your notebook, head the page **M5-15 The somatic and autonomic motor pathways, side by side** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can point at the one feature on the right-hand drawing that the left-hand drawing does not have at all, and say what follows from it.

Mini visual sheet

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

The whole nervous system on one page

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

2

One stimulus, all the way through

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

3

CNS against PNS, every term that changes

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

4

Where every cell body sits

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

5

From the scalp inward to the neuron

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

6

Cranial nerve nuclei by brainstem region

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

7

Six lesions and the deficits they name

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

8

CSF, choroid plexus back to the blood

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

Drawing pages

The drawings from Panel 2, with room to draw them. Brain dump first with everything closed, then draw, then correct yourself in a second color.

DRAWING 1

The brain, from memory, nothing open

DRAWING 2

A spinal cord segment with everything attached to it

DRAWING 3

The somatic and autonomic motor pathways, side by side

PART C

Lab Structure List.

Part C of the Module 5 packet. These are the lab structures and concepts to learn for the lab portion of the Module 5 exam. Treat it as the master list, and cross reference it against every lab worksheet. Be prepared to identify structures on models, images, the cadaver, and slides.

1. **Nervous Tissue, Histology**
2. **The Brain**
3. **The Brainstem and Cerebellum**
4. **Meninges, Ventricles, and CSF**
5. **The Spinal Cord**
6. **The Cranial Nerves**
7. **Peripheral Nerves and Plexuses**

CHAPTER 1

Nervous Tissue, Histology

THE SLIDES AND THE NEURON MODEL

Five slides, and her master list names all five. Fill in function and location as you go, because on this sheet the cell you are pointing at is often identified by the tissue it sits in rather than by its own shape.

☐ Multipolar neuron, on the slide and the model

FUNCTION ______________________ LOCATION ______________________

SPECIAL STRUCTURES ______________________

STRUCTURES TO IDENTIFY

- ☐ Cell body (soma)
- ☐ Nucleus
- ☐ Nucleolus
- ☐ Cell processes (dendrites)
- ☐ Axon
- ☐ Axon hillock
- ☐ Synaptic bulb (terminal end)
- ☐ Nissl bodies
- ☐ Neurofibrils

☐ Spinal cord, cross section

FUNCTION ______________________ LOCATION ______________________

SPECIAL STRUCTURES ______________________

STRUCTURES TO IDENTIFY

- ☐ White matter
- ☐ Gray matter
- ☐ Ventral (anterior) horn
- ☐ Dorsal (posterior) horn
- ☐ Lateral horn, if present
- ☐ Central canal

☐ Cerebral cortex

FUNCTION ______________________ LOCATION ______________________

SPECIAL STRUCTURES ______________________

STRUCTURES TO IDENTIFY

- ☐ Pyramidal cells
- ☐ Gray matter of the cortex
- ☐ Underlying white matter

☐ Cerebellum

FUNCTION ______________________ LOCATION ______________________

SPECIAL STRUCTURES ______________________

STRUCTURES TO IDENTIFY

- ☐ Purkinje cells
- ☐ Cerebellar cortex
- ☐ Arbor vitae

☐ Nerve, cross section

All of these are best seen on a cross section. Work outward from a single axon to the whole nerve.

FUNCTION ______________________ LOCATION ______________________

SPECIAL STRUCTURES ______________________

STRUCTURES TO IDENTIFY

- ☐ Axon
- ☐ Schwann cells
- ☐ Nodes of Ranvier
- ☐ Endoneurium
- ☐ Perineurium
- ☐ Epineurium
- ☐ Fascicle

☐ The neuroglia

Sort the six into the four of the central nervous system and the two of the peripheral nervous system, and give the job of each.

STRUCTURES TO IDENTIFY

- ☐ Astrocytes
- ☐ Oligodendrocytes
- ☐ Microglia
- ☐ Ependymal cells
- ☐ Schwann cells
- ☐ Satellite cells

A cluster of cell bodies is called a ______________________________ in the CNS and a ______________________________ in the PNS.

A bundle of axons is called a ______________________________ in the CNS and a ______________________________ in the PNS.

Which cell builds the myelin in the PNS, and which one builds it in the CNS?

____

CHAPTER 2

The Brain

THE BRAIN MODEL, THE SHEEP BRAIN, AND THE CADAVER

The largest identification block of the module. Work it surface first, then midsagittal, then inferior, because that is the order the models are set out in and it is how the tags follow.

☐ Surface features of the cerebrum

STRUCTURES TO IDENTIFY

- ☐ Gyrus
- ☐ Sulcus
- ☐ Longitudinal fissure
- ☐ Transverse fissure
- ☐ Precentral gyrus
- ☐ Precentral sulcus
- ☐ Central sulcus
- ☐ Postcentral gyrus
- ☐ Postcentral sulcus
- ☐ Lateral sulcus (Sylvian fissure)
- ☐ Insula

☐ The lobes

Two lobes are unpaired on her list and two are paired. Say which is which, and name the sulcus or fissure that bounds each lobe.

STRUCTURES TO IDENTIFY

- ☐ Frontal lobe
- ☐ Parietal lobe
- ☐ Temporal lobe
- ☐ Occipital lobe
- ☐ Insula

☐ Midsagittal section

STRUCTURES TO IDENTIFY

- ☐ Corpus callosum
- ☐ Sulcus of the corpus callosum
- ☐ Cingulate gyrus
- ☐ Cingulate sulcus
- ☐ Septum pellucidum
- ☐ Fornix
- ☐ Thalamus
- ☐ Massa intermedia (interthalamic adhesion)
- ☐ Hypothalamus
- ☐ Infundibulum
- ☐ Pineal gland
- ☐ Optic chiasm
- ☐ Cerebellum
- ☐ Transverse fissure

- [ ] **Inferior surface**

STRUCTURES TO IDENTIFY

- [ ] Uncus
- [ ] Parahippocampal gyrus
- [ ] Mammillary bodies
- [ ] Crus cerebri
- [ ] Optic chiasm
- [ ] Optic nerve
- [ ] Optic tract
- [ ] Olfactory bulb
- [ ] Olfactory tract

- [ ] **Functional areas of the cortex**

Name the gyrus each cortical area sits on, and say which hemisphere Broca's and Wernicke's areas are usually found in.

STRUCTURES TO IDENTIFY

- [ ] Primary motor cortex
- [ ] Primary sensory cortex
- [ ] Broca's area
- [ ] Wernicke's area
- [ ] Thalamus

- [ ] **The cerebellum**

STRUCTURES TO IDENTIFY

- [ ] Cerebellar hemispheres
- [ ] Vermis
- [ ] Folia
- [ ] Cerebellar cortex
- [ ] Arbor vitae
- [ ] Transverse fissure

- [ ] **The cerebral arterial circle (Circle of Willis)**

Trace the ring from a vertebral artery around to an anterior cerebral artery, naming every vessel on the way.

STRUCTURES TO IDENTIFY

- [ ] Internal carotid artery
- [ ] Anterior cerebral artery
- [ ] Anterior communicating artery
- [ ] Middle cerebral artery
- [ ] Posterior communicating artery
- [ ] Posterior cerebral artery
- [ ] Basilar artery
- [ ] Pontine arteries
- [ ] Vertebral arteries
- [ ] Superior cerebellar artery
- [ ] Anterior inferior cerebellar artery
- [ ] Posterior inferior cerebellar artery
- [ ] Anterior spinal artery

Which gyrus lies immediately anterior to the central sulcus, and what does it do?

____

CHAPTER 3

The Brainstem and Cerebellum

THE BRAIN MODEL AND THE SHEEP BRAIN

Three regions from the top down, plus the cerebellum hanging off the back of them. Every structure here is either a surface landmark or a stalk, so name the region before you name the part.

- [ ] **The three regions**

Name the cranial nerves that arise from each region.

STRUCTURES TO IDENTIFY

- [ ] Midbrain
- [ ] Pons
- [ ] Medulla oblongata
- [ ] Pontomedullary junction

- [ ] **The midbrain**

STRUCTURES TO IDENTIFY

- [ ] Cerebral peduncles
- [ ] Crus cerebri
- [ ] Cerebral aqueduct

☐ The pons

STRUCTURES TO IDENTIFY

- ☐ Basilar pons
- ☐ Middle cerebellar peduncle
- ☐ Fourth ventricle

☐ The medulla oblongata

Say what crosses at the decussation of the pyramids and what that crossing explains.

STRUCTURES TO IDENTIFY

- ☐ Pyramids
- ☐ Decussation of the pyramids
- ☐ Medullary olive
- ☐ Pontomedullary junction

☐ The cerebellum and its peduncles

Say which region of the brainstem each peduncle connects to.

STRUCTURES TO IDENTIFY

- ☐ Cerebellar hemispheres
- ☐ Vermis
- ☐ Folia
- ☐ Arbor vitae
- ☐ Superior cerebellar peduncle
- ☐ Middle cerebellar peduncle
- ☐ Inferior cerebellar peduncle

Which region of the brainstem is continuous with the spinal cord, and at what landmark?

____

CHAPTER 4

Meninges, Ventricles, and CSF

THE BRAIN MODEL AND THE CORONAL SECTION

Three membranes, three spaces, four ventricles, and one circuit. Her master list asks you to trace the fluid from the lateral ventricles to reabsorption, and it warns that some structures on that route are named but not seen on the model.

☐ The cranial meninges

Name the three layers in order from the skull inward.

STRUCTURES TO IDENTIFY

- ☐ Dura mater
- ☐ Periosteal layer of the dura
- ☐ Meningeal layer of the dura
- ☐ Arachnoid mater
- ☐ Pia mater

☐ The dural extensions

For each one, say where it is found and which structures it separates.

STRUCTURES TO IDENTIFY

- ☐ Falx cerebri
- ☐ Falx cerebelli
- ☐ Tentorium cerebelli

☐ The dural venous sinuses

Say where the sigmoid sinus leaves the skull and what the blood becomes once it does.

STRUCTURES TO IDENTIFY

- ☐ Superior sagittal sinus
- ☐ Inferior sagittal sinus
- ☐ Straight sinus
- ☐ Confluence of sinuses
- ☐ Transverse sinus
- ☐ Sigmoid sinus

- [ ] **The meningeal spaces**

Say which of these spaces holds cerebrospinal fluid and which is only a potential space.

STRUCTURES TO IDENTIFY

- [ ] Epidural space
- [ ] Subarachnoid space
- [ ] Subdural space
- [ ] Arachnoid granulations

- [ ] **The ventricles**

STRUCTURES TO IDENTIFY

- [ ] Lateral ventricle
- [ ] Interventricular foramen of Monro
- [ ] Third ventricle
- [ ] Cerebral aqueduct
- [ ] Fourth ventricle
- [ ] Central canal
- [ ] Septum pellucidum

- [ ] **The choroid plexus, on the slide and the model**

FUNCTION ______________________ LOCATION ______________________

SPECIAL STRUCTURES ______________________

STRUCTURES TO IDENTIFY

- [ ] Choroid plexus
- [ ] Ependymal cells
- [ ] Capillary network

Trace cerebrospinal fluid from the lateral ventricles to its reabsorption, naming every structure it passes. Include the medial and lateral apertures and the central canal even though they are not seen on the model.

____

____

CHAPTER 5

The Spinal Cord

THE CORD MODEL, THE CROSS SECTION, AND THE CADAVER

The cord in two views. The whole cord from the outside gives you the enlargements and the terminations; the cross section gives you the horns, the commissure, and the canal.

- [ ] **The spinal meninges**

Say what fills the epidural space in the vertebral canal, and why there is no epidural space inside the cranium.

STRUCTURES TO IDENTIFY

- [ ] Dura mater
- [ ] Arachnoid mater
- [ ] Pia mater
- [ ] Epidural space
- [ ] Subdural space
- [ ] Subarachnoid space
- [ ] Denticulate ligaments

- [ ] **External anatomy of the cord**

Give the vertebral level where the cord ends, and say what fills the canal below that point.

STRUCTURES TO IDENTIFY

- [ ] Cervical enlargement
- [ ] Lumbar enlargement
- [ ] Conus medullaris
- [ ] Cauda equina
- [ ] Filum terminale
- [ ] Anterior median fissure
- [ ] Posterior median sulcus

- [ ] **The cord in cross section**

Say which segments of the cord carry a lateral horn and what sits in it.

FUNCTION ______________________ LOCATION ______________________

SPECIAL STRUCTURES ______________________

STRUCTURES TO IDENTIFY

- [ ] Ventral (anterior) horn
- [ ] Lateral horn
- [ ] Dorsal (posterior) horn
- [ ] Gray matter
- [ ] White matter
- [ ] Gray commissure
- [ ] Central canal
- [ ] Anterior white commissure
- [ ] Anterior column
- [ ] Lateral column
- [ ] Posterior column

Where in the vertebral column is a lumbar puncture performed, and why is that level safe?

____

CHAPTER 6

The Cranial Nerves

THE BRAIN MODEL, THE SKULL, AND THE SHEEP BRAIN

Twelve nerves, and her master list asks four things of each one: identify it, classify it as sensory, motor, or both, give its function, and find the opening it leaves the skull through. Do all four every time you point at one.

- [ ] **The twelve cranial nerves, I to XII**

For each nerve give the number, the name, the classification as sensory, motor, or both, and the function.

STRUCTURES TO IDENTIFY

- [ ] CN I, Olfactory
- [ ] CN II, Optic
- [ ] CN III, Oculomotor
- [ ] CN IV, Trochlear
- [ ] CN V, Trigeminal
- [ ] CN VI, Abducens
- [ ] CN VII, Facial
- [ ] CN VIII, Vestibulocochlear
- [ ] CN IX, Glossopharyngeal
- [ ] CN X, Vagus
- [ ] CN XI, Accessory
- [ ] CN XII, Hypoglossal

- [ ] **Structures associated with the cranial nerves**

Name the nerve that passes through each opening.

STRUCTURES TO IDENTIFY

- [ ] Pontomedullary junction
- [ ] Internal acoustic meatus
- [ ] Olfactory bulb
- [ ] Olfactory tract
- [ ] Olfactory foramina in the cribriform plate
- [ ] Optic nerve
- [ ] Optic chiasm
- [ ] Optic tract
- [ ] Superior orbital fissure
- [ ] Optic canal

- [ ] **The branches of the trigeminal nerve**

Name the skull opening each division passes through.

STRUCTURES TO IDENTIFY

- [ ] Ophthalmic division (V1)
- [ ] Maxillary division (V2)
- [ ] Mandibular division (V3)

☐ **The extrinsic eye muscles**

Name the cranial nerve that supplies each muscle. Three nerves cover all seven.

STRUCTURES TO IDENTIFY

- ☐ Lateral rectus
- ☐ Medial rectus
- ☐ Superior rectus
- ☐ Inferior rectus
- ☐ Superior oblique
- ☐ Inferior oblique
- ☐ Levator palpebrae superioris

Which three cranial nerves are purely sensory? ______________________________

Which four cranial nerves are mixed? ______________________________

CHAPTER 7

Peripheral Nerves and Plexuses

THE CORD MODEL, THE CADAVER, AND THE TORSO

The spinal nerve is the hinge of this sheet. Everything before it is a root, everything after it is a ramus, and her list asks you to say what kind of information travels in each.

☐ **The spinal nerve and its roots**

For each structure say what it carries: sensory axons, motor axons, mixed, or sensory cell bodies.

STRUCTURES TO IDENTIFY

- ☐ Ventral (anterior) root
- ☐ Dorsal (posterior) root
- ☐ Dorsal root ganglion
- ☐ Spinal nerve
- ☐ Intervertebral foramen
- ☐ Rootlets

☐ **The rami and the sympathetic connections**

Say which ramus forms the plexuses and which one serves the deep muscles and skin of the back.

STRUCTURES TO IDENTIFY

- ☐ Ventral (anterior) ramus
- ☐ Dorsal (posterior) ramus
- ☐ Gray ramus communicans
- ☐ White ramus communicans
- ☐ Sympathetic chain ganglion
- ☐ Sympathetic trunk

☐ **The plexuses**

Give the spinal nerves that form each plexus and the region of the body each one serves.

STRUCTURES TO IDENTIFY

- ☐ Cervical plexus
- ☐ Brachial plexus
- ☐ Lumbar plexus
- ☐ Sacral plexus

☐ **Nerves to find on the cadaver and the models**

Name the plexus the phrenic nerve comes from and the structure it supplies.

STRUCTURES TO IDENTIFY

- ☐ Phrenic nerve
- ☐ Intercostal nerves

Which root carries a ganglion, and whose cell bodies are in it? ______________________________

A patient loses sensation and strength across one whole upper limb. Which plexus is damaged?

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